

# Trade Transparency and Buyer Learning

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## Abstract

This paper studies the welfare effects of trade transparency in a dynamic adverse-selection market where buyers can acquire costly, noisy private information about seller quality. Under transparency, buyers observe past trade occurrences but not transaction prices. This public history creates two competing forces: a learning effect, as later buyers use past trading activity to economize on costly information acquisition, and a strategic distortion effect, as sellers alter current trading decisions to influence future beliefs. The welfare effect of transparency depends on the balance of these forces: it increases welfare when the informational gains and savings in acquisition costs outweigh the distortion of trade, but reduces welfare when reliance on public history discourages valuable private information acquisition and seller manipulation generates sufficiently large allocative losses.

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# 1 Introduction

In many markets, sellers repeatedly transact with changing sets of buyers rather than forming long-term relationships with the same customers. Online labor markets provide a prominent example: Pallais (2014) documents that repeat interactions between workers and the same employers on oDesk are not the norm, while Horton and Chilton (2010) describe crowd-sourcing markets as organized around small, short-duration tasks. Similar patterns arise in markets involving contractors, real estate agents, travel agents, and sellers of artisanal or specialized goods. In such environments, buyers face uncertainty about seller quality but have limited opportunities to learn through repeated personal experience.

Buyers may instead acquire costly private signals about seller quality through interviews, background checks, inspections, trial tasks, or personal research. Although future buyers do not observe these signals directly, they may learn from the trading decisions those signals induce when past trading activity is publicly observable. By allowing later buyers to learn from earlier market activity, transparency can reduce their incentive to acquire an independent private signal and thereby save duplicative information-acquisition costs. At the same time, making trade histories observable gives sellers an incentive to alter current trading decisions in order to influence future beliefs about their quality. This paper asks whether, once these responses are taken into account, making trade histories public increases social surplus relative to an opaque market.

I study a form of transparency under which later buyers observe whether the seller traded in the previous period but not the transaction price or other details. I refer to this form of observability as *trade transparency* and, hereafter, simply as *transparency*. For example, online labor platforms may reveal completed-job histories, and home-service platforms may display how many times a provider has been hired, without revealing the prices of those transactions.

To analyze this environment, I develop a two-period model in which a long-lived seller has a fixed, privately known quality, either high or low, and interacts with one short-lived

buyer in each period. Each buyer is initially uncertain about the seller’s type and may pay a cost to acquire a noisy private signal before making a take-it-or-leave-it offer. I compare transparency with an opaque benchmark in which Buyer 2 does not observe the period-1 trading outcome.

Transparency affects welfare through two equilibrium channels. The first is a *learning effect*. When period-1 trade is observable, Buyer 2 can infer information about seller quality from the trading history generated by Buyer 1’s private signal and the seller’s response. This changes Buyer 2’s incentive to acquire his own private signal. The welfare effect is ambiguous: relying on the public history can save the cost of acquiring information that is no longer needed, but it can also discourage private information acquisition that would have supported more efficient trade.

The second is a *strategic distortion effect*. Because the seller’s period-1 trading decision affects future beliefs, the low-quality seller may depart from her static trading incentives in order to influence Buyer 2’s beliefs about her quality. This distortion is unambiguously welfare-reducing because it moves period-1 trade away from the surplus-maximizing allocation. By changing the informativeness of the public history, the seller’s distortion also feeds back into Buyer 2’s information-acquisition decision.

The form of strategic distortion depends on whether trade with the low-quality seller is socially efficient. When such trade is inefficient, the low-quality seller may accept an offer she would otherwise reject in order to generate the trading outcome associated with more favorable future beliefs. Transparency can therefore induce *overtrading*: trade occurs even though it is socially inefficient. When trade with the low-quality seller is socially efficient, she may instead reject an offer she would otherwise accept in order to generate the history associated with more favorable future beliefs. Transparency can then induce *undertrading*: efficient trade fails to occur. Thus, the same reputational motive can generate opposite distortions depending on the social value of low-quality trade, both of which reduce welfare.

Transparency can strictly increase welfare when the gains from the learning effect are

sufficiently large. Even when the low-quality seller distorts her period-1 trading decision, Buyer 2 may optimally forgo acquiring his own private signal while still engaging in more efficient trade than in the opaque market. The resulting savings in information-acquisition costs and gains from efficient trade outweigh the welfare loss from seller distortion.

Transparency can also strictly reduce welfare when reliance on the public trading history discourages information acquisition that would otherwise support efficient trade. When Buyer 2 forgoes his own private signal, he may therefore make his period-2 trading decision using less accurate information about seller quality. The public history may be misleading either because Buyer 1 received an inaccurate signal or because the low-quality seller distorted her period-1 trading decision to influence future beliefs. The resulting loss from foregone efficient trade, together with the welfare loss from seller distortion, can outweigh the savings in information-acquisition costs.

The analysis also shows that, in some cases, transparent markets can perform worse as buyers' private signals become more precise. Holding all other primitives fixed, an arbitrarily small increase in signal precision can induce a discrete change in equilibrium behavior, causing the transparent market to go from outperforming the opaque market to performing worse in terms of total welfare. Greater signal precision changes both the information conveyed by the public history and the seller's incentive to distort her period-1 trading decision, which can trigger this equilibrium shift.

The core mechanism is robust to alternative market environments. In particular, it survives extensions with stochastic buyer arrival and noisy observability of past trade. These results highlight a limitation of market transparency: when sellers distort current trading decisions to influence future beliefs, transparency can alter both trade and information acquisition in ways that reduce welfare.

The remainder of this section reviews the related literature. Section 2 presents the model, Section 3 analyzes the opaque benchmark, Section 4 studies trade transparency, Section 5 provides additional analyses and extensions, and Section 6 concludes.

**Related Literature.** This paper studies a dynamic market in which a long-lived privately informed seller repeatedly meets short-lived buyers. The main question is how transparency of past trading activity affects market outcomes when buyers can acquire costly private signals about seller quality. The paper connects to work on transparency in dynamic markets, private buyer signals about seller or asset quality, and endogenous information acquisition before trade.

Most closely related on the transparency and repeated-trade dimensions is Kaya and Roy (2024), who study the observability of past trading activity in a market where an informed seller repeatedly meets short-lived buyers. They show how the welfare effects of transparency depend on buyer competition and initial beliefs through market screening. Related work by Kaya and Roy (2022*a*) and Kaya and Roy (2022*b*) examines other dimensions of transparency: the former varies the length of the observable trading record, while the latter asks whether transaction prices should be observable in addition to past trading activity. My paper also studies the observability of past trading activity, focusing on whether Buyer 2 observes whether period-1 trade occurred, but allows buyers to endogenously acquire a costly private signal about seller quality. Transparency therefore also affects buyers' incentive to become privately informed.

Another close comparison is Aghamolla and Matsuno (2026), who study disclosure of past trade in a repeated-trade environment where buyers also receive private signals about seller quality. Their endogenous information margin is the seller's disclosure decision, whereas buyers' private signals are exogenous. In my model, the observability of past trade is fixed by the market environment, and buyers instead decide whether to acquire a costly private signal.

A number of related papers study how observable market history affects learning and trading dynamics in single-sale adverse-selection environments (Kaya and Liu, 2015; Kim, 2017; Hwang, 2018). These papers differ in the market history that is observed: Kaya and Liu (2015) focus on past rejected offers, while Kim (2017) and Hwang (2018) study time-on-

market observability. Hwang (2018) differs further in that the seller is initially uninformed about the quality of her good and becomes privately informed over time. Because these are single-sale environments, the interaction ends once the asset is sold. In my setting, the seller can trade repeatedly, so a completed trade can itself become part of the history observed by a subsequent buyer.

Within this literature, Kaya and Kim (2018) and Aghamolla and Hashimoto (2023) additionally allow buyers to receive private signals about asset quality. Kaya and Kim (2018) make time on the market observable, whereas Aghamolla and Hashimoto (2023) make the arrival of buyers' one-time offers publicly observable while keeping offer values private, allowing buyers to learn from the timing of others' actions. In both papers, buyers' private signals are exogenously given, whereas in my model buyers choose whether to acquire a costly private signal.

There is also a literature on transparency in financial markets, spanning different trading institutions and informational frictions. For example, Fuchs, Öry and Skrzypacz (2016) study a privately informed asset owner who experiences a liquidity shock and compare public and private past price offers in a dynamic adverse-selection market. Asriyan, Fuchs and Green (2017) study information spillovers across markets for correlated assets, where transactions in one market reveal information about asset values in others. Vairo and Dworczak (2023) study the design of pre- and post-trade transparency in over-the-counter markets. In these examples, buyers' private information acquisition is not an endogenous margin.

There is also work on endogenous information acquisition before trade in static environments. Ravid, Roesler and Szentes (2022) study costly learning about the buyer's own valuation, while Jang and Kang (2021) study costly buyer information acquisition about an informed seller's asset quality. My paper studies how this acquisition decision interacts with publicly observable trading history in a dynamic market.

A separate line of work studies repeated trade with imperfect information about past transactions. Dilmé (2025) considers an independent-values environment with a long-lived

privately informed buyer facing a sequence of short-lived sellers who observe noisy information about his past purchases. In my setting, the long-lived informed player is instead the seller, and short-lived buyers can endogenously acquire information about her quality.

## 2 Model

**Preliminaries.** I consider a two-period model,  $t \in \{1, 2\}$ , with a long-lived seller who can supply at most one unit in each period. The seller has a privately known type  $\theta \in \{H, L\}$ , where  $H$  denotes high quality and  $L$  denotes low quality. Her type is fixed across periods and determines her cost of producing one unit,  $c_\theta$ , where  $c_H > c_L$ . The seller discounts future payoffs by  $\delta \in (0, 1)$ .

In each period, a short-lived buyer enters the market. The two buyers are ex ante identical and uninformed about the seller's quality. They share a common prior  $\mu_0 = \Pr(\theta = H)$  that the seller is of high quality. A buyer's valuation depends on the seller's type: a unit supplied by a type- $\theta$  seller is worth  $v_\theta$ , where  $v_H > v_L$ .

I impose the following ordering on the model parameters:  $v_H > c_H > \max\{c_L, v_L\} \geq 0$ . Thus, a unit supplied by a high-quality seller is more valuable to the buyer, and the high-quality seller has a higher production cost than the low-quality seller. Since  $v_H > c_H$ , there are always positive gains from trade with a high-quality seller. I study both the case in which  $v_L > c_L$ , so that trade with a low-quality seller also generates positive gains, and the case in which  $v_L < c_L$ , so that trade with a low-quality seller generates losses.

**Information acquisition and timing.** In each period, the buyer can learn more about the seller's quality by acquiring a private signal  $s \in \{h, l\}$  at cost  $\kappa$ . The signal is binary and has accuracy  $\alpha > \frac{1}{2}$ , where  $\Pr(s = h \mid \theta = H) = \Pr(s = l \mid \theta = L) = \alpha$ . I consider two types of markets: opaque and transparent. In an opaque market, Buyer 2 observes no information about period 1. In a transparent market, Buyer 2 observes whether or not trade occurred in period 1, but does not observe Buyer 1's offer. In each period, after making

his signal-acquisition decision, the buyer makes a take-it-or-leave-it offer to the seller. If he acquires a signal, he conditions his offer on the signal realization. The seller then accepts or rejects the offer.

**Histories and player strategies.** Let  $\mathcal{H}_i$  denote the set of histories observed by Buyer  $i$  upon entering the market. Buyer 1 observes no prior history, so  $\mathcal{H}_1 = \emptyset$ . Buyer 2 also has  $\mathcal{H}_2 = \emptyset$  under opacity, while under transparency  $\mathcal{H}_2 = \{t, nt\}$ , where  $t$  denotes that trade occurred in period 1 and  $nt$  denotes that no trade occurred. After observing the relevant history, Buyer  $i$  chooses whether to acquire information. His acquisition strategy is  $d_i : \mathcal{H}_i \rightarrow \Delta(a, n)$ , where  $a$  denotes acquisition and  $n$  denotes no acquisition. Let  $\mathcal{I} = \{n, h, l\}$  denote a buyer's possible information states at the offer stage, where  $n$  denotes no signal acquisition and  $h$  and  $l$  denote the two possible signal realizations. Buyer  $i$ 's offer strategy is  $p_i : \mathcal{H}_i \times \mathcal{I} \rightarrow \Delta(\mathbb{R}_+)$ .

For each type  $\theta \in \{H, L\}$ , the seller's strategy specifies an acceptance probability after every feasible history and current offer. Let  $\sigma_i^\theta(h_i, p) \in [0, 1]$  denote the probability that a type- $\theta$  seller accepts offer  $p$  in period  $i$  following history  $h_i$ .

**Beliefs and equilibrium.** Each buyer holds a posterior belief about the seller's type conditional on the information available to him. I consider perfect Bayesian equilibria, in which strategies are sequentially optimal given beliefs and beliefs are updated according to Bayes' rule whenever possible.

**Welfare.** I measure welfare by ex-ante expected social surplus, defined as the discounted sum of gains from trade across the two periods, net of information-acquisition costs. Period-2 surplus is discounted by  $\delta$ .

**Static benchmark without access to information acquisition.** To characterize the opaque-market equilibrium, first consider a single-period market in which the buyer cannot

acquire a private signal and therefore conditions his offer solely on the prior  $\mu_0$ . A type- $\theta$  seller accepts an offer  $p$  if and only if  $p \geq c_\theta$ . The buyer can trade with both types by offering  $c_H$ . His alternative is to offer  $c_L$  when  $v_L > c_L$ , trading only with the low-quality seller, or an amount below  $c_L$  when  $v_L < c_L$ , inducing no trade. The cutoff belief at which he is indifferent is

$$\hat{\mu} = \begin{cases} \frac{c_H - c_L}{v_H - c_L}, & v_L > c_L, \\ \frac{c_H - v_L}{v_H - v_L}, & v_L < c_L. \end{cases} \quad (1)$$

Thus, the buyer offers  $c_H$  if and only if  $\mu_0 \geq \hat{\mu}$ .

Finally, I focus on parameter regions where at least one buyer chooses to acquire information in either the transparent or opaque market, as these are the regions in which transparency affects learning. Outside these regions, no buyer acquires information in either market, and I do not analyze those cases further.

### 3 Opaque Market

In the opaque market, Buyer 2 observes no information about period-1 trade. Hence, the seller's period-1 action does not affect her continuation payoff, and the market outcome is the period-by-period repetition of the static outcome. Each buyer first decides whether to acquire a signal at cost  $\kappa$  and then makes a take-it-or-leave-it offer. Because Buyer 2 does not observe period-1 trade, both buyers have the same information, prior, and valuations, and therefore face the same acquisition problem. Appendix B shows that there exists a non-degenerate range of priors over which information is acquired if and only if

$$\kappa < (2\alpha - 1)\hat{\mu}(v_H - c_H) \iff \alpha > \frac{1}{2} + \frac{\kappa}{2\hat{\mu}(v_H - c_H)}. \quad (2)$$

When this condition holds, there exist lower and upper bounds on the prior such that buyers acquire the signal if and only if  $\mu_0$  lies between them. For  $v_L < c_L$ , these bounds are

$\underline{\mu} < \hat{\mu} < \bar{\mu}$ ; for  $v_L > c_L$ , they are  $\underline{\mu} < \hat{\mu} < \tilde{\mu}$ . When (2) holds, there exist lower and upper bounds on the prior such that buyers acquire the signal if and only if  $\mu_0$  lies between them. For  $v_L < c_L$ , these bounds are  $\underline{\mu} < \hat{\mu} < \bar{\mu}$ ; for  $v_L > c_L$ , they are  $\underline{\mu} < \hat{\mu} < \tilde{\mu}$ . Define the acquisition region by

$$\mathcal{A} \equiv \begin{cases} (\underline{\mu}, \bar{\mu}), & \text{if } v_L < c_L, \\ (\underline{\mu}, \tilde{\mu}), & \text{if } v_L > c_L. \end{cases} \quad (3)$$

In general,  $\mathcal{A}$  is asymmetric around  $\hat{\mu}$ . Holding the prior fixed, the  $\mathcal{A}$  expands as signal precision  $\alpha$  increases or the acquisition cost  $\kappa$  decreases. The expression for the bounds are given in Appendix B.

**Proposition 1.** *Choose any  $\underline{p} < c_L$  if  $v_L < c_L$ , and set  $\underline{p} = c_L$  otherwise. Then, in an opaque market, for  $i \in \{1, 2\}$ :*

- (i) *The seller accepts an offer if and only if it is weakly above her cost.*
- (ii) *When (2) holds, Buyer  $i$  acquires the signal if and only if  $\mu_0 \in \mathcal{A}$ . Conditional on acquisition, Buyer  $i$  offers  $c_H$  following a high signal and  $\underline{p}$  following a low signal. If  $v_L < c_L$ , trade occurs if and only if the signal is high; if  $v_L > c_L$ , trade occurs unless the seller is high quality and the signal is low.*
- (iii) *When (2) does not hold, Buyer  $i$  does not acquire the signal and offers  $c_H$  if  $\mu_0 \geq \hat{\mu}$  and  $\underline{p}$  if  $\mu_0 < \hat{\mu}$ .*

The proof is in Appendix B. Conditional on acquisition, the resulting trade pattern determines the informational content of period-1 trade. If  $v_L < c_L$ , trade occurs if and only if the signal is high, so the trade outcome fully reveals the signal. If  $v_L > c_L$ , no trade occurs only when the seller is high quality and the signal is low, and therefore reveals high quality. Without acquisition, buyers condition only on the prior. For  $v_L < c_L$ , trade occurs if and only if  $\mu_0 \geq \hat{\mu}$ . For  $v_L > c_L$ , both types trade if  $\mu_0 \geq \hat{\mu}$ , but only the low-quality seller trades if  $\mu_0 < \hat{\mu}$ . Given the equilibrium in Proposition 1, the expected total surplus when

$\mu_0 \in \mathcal{A}$  is given by

$$ETS_{\text{opaque}}^a = \begin{cases} (1 + \delta) [\mu_0 \alpha (v_H - c_H) + (1 - \mu_0)(v_L - c_L) - \kappa], & \text{if } v_L > c_L, \\ (1 + \delta) [\mu_0 \alpha (v_H - c_H) + (1 - \mu_0)(1 - \alpha)(v_L - c_L) - \kappa], & \text{if } v_L < c_L. \end{cases}$$

On the other hand, when  $\mu_0 \notin \mathcal{A}$

$$ETS_{\text{opaque}}^n = \begin{cases} (1 + \delta) [\mu_0 (v_H - c_H) + (1 - \mu_0)(v_L - c_L)], & \text{if } \mu_0 \geq \hat{\mu}, \\ (1 + \delta)(1 - \mu_0) [v_L - \min\{v_L, c_L\}], & \text{if } \mu_0 < \hat{\mu}. \end{cases}$$

## 4 Transparent Market

I now analyze equilibrium in the transparent market. The key difference from the opaque benchmark is that Buyer 2 observes the period-1 trade outcome before making his information-acquisition and offer decisions. The period-1 outcome can therefore convey information about Buyer 1's private signal and affect Buyer 2's beliefs about seller quality. Anticipating this, the seller may adjust her period-1 reservation price because her current trading decision affects her continuation payoff.

Transparency affects welfare relative to the opaque benchmark through two mechanisms introduced earlier. First, the *learning effect* arises because the period-1 trade outcome affects Buyer 2's information-acquisition decision. This effect has two components: savings in information-acquisition costs and changes in period-2 trade when Buyer 2 forgoes private learning. The first component raises welfare, while the second may raise or lower welfare depending on how trade changes relative to the opaque benchmark. The overall learning effect is therefore ambiguous.

Second, the *strategic distortion effect* arises because a low-quality seller may alter her period-1 trading behavior to improve her continuation payoff. Unlike the learning effect, strategic distortion is unambiguously welfare-reducing. As shown below, it takes the form

of inefficient overtrading when  $v_L < c_L$  and inefficient undertrading when  $v_L > c_L$ .

I first characterize equilibrium and welfare when there are gains from trading only with the high-quality seller, i.e.,  $v_L < c_L$ .

#### 4.1 Gains from trade only with the high-quality seller

Suppose that  $v_H > c_H > c_L > v_L$ , so that gains from trade arise only with the high-quality seller. I first establish that transparency does not affect Buyer 1's incentive to acquire information compared to the opaque benchmark.

**Lemma 4.1** (Invariance of Buyer 1's information acquisition). *Generically, Buyer 1 acquires his private signal in the transparent market if and only if he acquires it in the opaque market.*

The proof is provided in Appendix C. The key argument is that the low-quality seller's equilibrium period-1 reservation price is at least  $v_L$ . Buyer 1 therefore cannot earn a positive profit from trading with the low-quality seller, so transparency leaves his information-acquisition incentive unchanged.

Define  $\bar{\delta} := (c_L - v_L)/(c_H - c_L)$ ,  $\tilde{\alpha} := (c_H - 2c_L + v_L)/(c_H - c_L)$ , and  $\hat{\alpha} := (c_H - v_L)/(c_H + c_L - 2v_L)$ . Recall from (3) that  $\underline{\mu}$  and  $\bar{\mu}$  denote the bounds of the acquisition region in the opaque market, and let  $\mu^*$ ,  $\mu^\dagger$ , and  $\mu^\ddagger$  denote the belief thresholds defined in (A.2) in Appendix C. For sufficiently small  $\kappa$ , their ordering depends on signal accuracy: if  $\alpha < \hat{\alpha}$ , then  $\mu^* < \underline{\mu} < \mu^\dagger < \mu^\ddagger < \bar{\mu}$ , whereas if  $\alpha > \hat{\alpha}$ , then  $\underline{\mu} < \mu^* < \mu^\dagger < \mu^\ddagger < \bar{\mu}$ .

**Theorem 1** (Transparency, information acquisition, and social surplus). *Generically, in any equilibrium in which Buyer 1 acquires information in the transparent market:*

- (I) *On the equilibrium path, Buyer 2's probability of acquiring a private signal is lower in the transparent market than in the opaque benchmark.*
- (II) *There exists  $\kappa^* > 0$  such that, if the cost of private signal acquisition  $\kappa < \kappa^*$ , the effect of market transparency on social surplus relative to an opaque benchmark is as follows:*

- (A) *If beliefs are optimistic, in particular, if  $\mu_0 \in (\mu^\dagger, \bar{\mu})$ : transparency is welfare improving.*
- (B) *If beliefs are pessimistic, in particular, if  $\mu_0 \in (\underline{\mu}, \mu^\dagger)$ :*
- (i) *If  $\alpha < \hat{\alpha}$ , transparency is welfare reducing.*
  - (ii) *If  $\alpha > \hat{\alpha}$ , transparency is welfare improving for  $\mu_0 \in (\underline{\mu}, \mu^*)$ , and welfare reducing for  $\mu_0 \in (\mu^*, \mu^\dagger)$ .*
- (C) *If beliefs are intermediate, in particular, if  $\mu_0 \in (\mu^\dagger, \mu^\ddagger)$ :*
- (i) *If  $\delta < \bar{\delta}$ , transparency is welfare improving.*
  - (ii) *If  $\delta > \bar{\delta}/(1 - \alpha)$  transparency is welfare reducing.*

The first part of Theorem 1 concerns Buyer 2's information acquisition. In the transparent market, his acquisition decision varies with the observed period-1 trade outcome. On the equilibrium path, he acquires a private signal with lower probability than in the opaque market because the observed trade outcome partially substitutes for private information.

The second part of the theorem establishes that, for sufficiently small acquisition costs—formally, for  $\kappa < \kappa^*$ , with  $\kappa^* > 0$  defined in Appendix C—the welfare effect of transparency depends jointly on the buyers' prior belief, signal precision, and the seller's patience, as measured by her discount factor.

Recall that transparency affects welfare through the learning and strategic distortion effects. The learning effect saves information-acquisition costs, while its trade component may reinforce or offset these savings. In the present case,  $v_L < c_L$ , so more trade with the high-quality seller raises welfare, whereas more trade with the low-quality seller lowers it. The strategic distortion effect operates through strategic overtrading by the low-quality seller, who may accept a below-cost period-1 offer to influence Buyer 2's future beliefs. Figure 1 summarizes the second part of the theorem.

Under optimistic beliefs, observing period-1 trade is sufficiently informative about the seller being high quality that Buyer 2 has less incentive to acquire a costly private signal

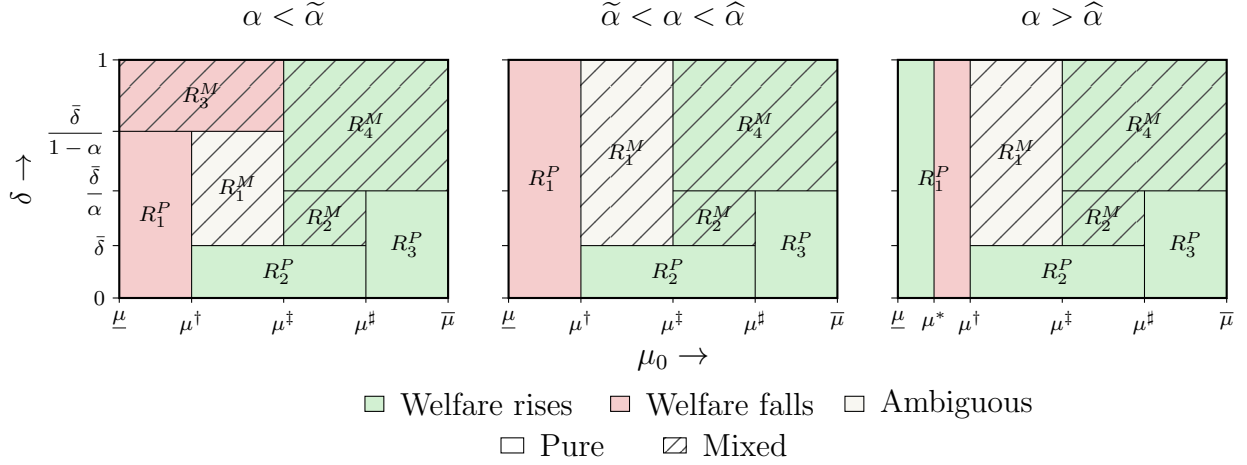


Figure 1: Pure- and mixed-strategy equilibrium regions and the corresponding welfare effects of transparency in the  $(\mu_0, \delta)$ -space. Hatched regions denote mixed-strategy equilibria. The panels show how equilibrium configurations and welfare effects vary with  $\alpha$ .

and is likely to make an offer accepted by both seller types. The learning effect is therefore favorable, through both acquisition-cost savings and more trade with the high-quality seller. These gains outweigh the welfare losses from any strategic overtrading by the low-quality seller in period 1 and any additional inefficient trade with her in period 2.

Under pessimistic beliefs, Buyer 2 interprets observing no trade as particularly unfavorable news about seller quality and is therefore likely to make an offer rejected by both seller types. In this region, the welfare effect depends on signal precision ( $\alpha$ ) and seller patience ( $\delta$ ). Greater seller patience strengthens the low-quality seller's incentive to distort, whereas greater signal precision weakens it by making her more likely to be identified when Buyer 2 acquires his private signal.

When  $\alpha < \hat{\alpha}$ , transparency reduces welfare. Lower signal precision weakens the ability to identify the low-quality seller, which strengthens her incentive to distort. In mixed-strategy equilibria, this distortion, together with the resulting changes in period-2 trade, outweighs the savings in information-acquisition costs. Even in pure-strategy equilibria, where strategic distortion is absent, the learning effect remains negative because the loss of efficient trade with the high-quality seller exceeds the acquisition-cost savings. When  $\alpha > \hat{\alpha}$ , sufficiently precise signals eliminate the low-quality seller's incentive to distort. For sufficiently

pessimistic beliefs, the learning effect raises welfare: Buyer 2 saves information-acquisition costs, and the induced reduction in trade lowers inefficient trade with the low-quality seller. As beliefs become less pessimistic, however, private learning becomes more valuable because it can reveal a high-quality seller and induce efficient trade. The resulting welfare loss from foregone trade can then outweigh the acquisition-cost savings, so transparency lowers welfare.

For intermediate beliefs, the welfare effect depends on seller patience. When  $\delta < \bar{\delta}$ , the low-quality seller does not distort, and the savings from reduced private-signal acquisition dominate, so transparency raises welfare. For  $\bar{\delta} < \delta < \bar{\delta}/(1-\alpha)$ , the welfare effect is generally ambiguous, and additional parameter restrictions are needed to determine its sign. When  $\delta > \bar{\delta}/(1-\alpha)$ , strategic distortion dominates and transparency lowers welfare. This last case arises only when  $\alpha < \tilde{\alpha}$ .

The pure- and mixed-strategy analyses below characterize these equilibrium responses and show how they jointly deliver the welfare regions in Theorem 1. I first characterize the pure-strategy equilibria, which arise when the low-quality seller is sufficiently impatient, and then turn to the mixed-strategy equilibria that arise when she is sufficiently patient. For each class of equilibria, I characterize Buyer 2's history-contingent information-acquisition behavior and derive the associated welfare implications relative to the opaque benchmark.

#### 4.1.1 Pure-strategy equilibria

In the pure-strategy equilibria, Buyer 1 acquires his private signal in period 1 and conditions his offer on its realization. After a high signal, he offers  $c_H$ , which both seller types accept; after a low signal, he offers some  $\tilde{p} < v_L$ , which both types reject. In particular, the low-quality seller does not distort her period-1 acceptance behavior. Thus, on the equilibrium path, trade reveals a high signal and no trade a low signal. Buyer 2's information-acquisition decision then depends on how this observed trade outcome shifts his beliefs relative to the prior: he may acquire only after trade, only after no trade, or not at all.

Transparency therefore affects welfare only through the learning effect. Relative to the

opaque benchmark, the observed trade outcome reduces Buyer 2's private signal acquisition, saving information-acquisition costs, while the associated change in period-2 trade may either reinforce or offset these savings.

**Equilibrium behavior.** (1) illustrates that pure-strategy equilibria arise in three regions— $R_1^P$ ,  $R_2^P$ , and  $R_3^P$ —of the  $(\mu_0, \delta)$ -space. Let  $\mu^\sharp$  denote the additional belief cutoff defined in Equation (A.2) in Appendix C, with  $\mu^\dagger < \mu^\sharp < \bar{\mu}$ .

Buyer 2's equilibrium behavior differs across these regions. In  $R_1^P$ , buyers' initial belief is pessimistic. On the equilibrium path, observing no trade in period 1 reveals a low signal realization and further lowers Buyer 2's belief that the seller is high quality, so that acquiring his private signal is no longer worthwhile. In contrast, observing trade in period 1 reveals a high signal realization and improves Buyer 2's belief, but leaves it in the region where signal acquisition remains valuable. Buyer 2 therefore acquires his private signal only after observing trade in period 1; after observing no trade, he does not acquire and instead offers some  $p < c_L$ .

In  $R_2^P$ , buyers' initial belief is intermediate. Observing trade in period 1 makes Buyer 2 sufficiently optimistic, while observing no trade makes him sufficiently pessimistic, so that private signal acquisition is not worthwhile after either history. He therefore never acquires his private signal, offering  $c_H$  after observing trade in period 1 and some  $p < c_L$  after observing no trade in period 1.

In  $R_3^P$ , Buyer 2 acquires his private signal only after observing no trade in period 1; after observing trade, he does not acquire and instead offers  $c_H$ . The equilibrium strategies of the remaining players are common across the three regions and are specified in Appendix C. To illustrate how the incentive conditions underlying these regions arise, consider  $R_3^P$  as a representative case.

The low-quality seller rejects the low period-1 offer because her acceptance decision affects Buyer 2's subsequent inference and hence her continuation payoff. Given Buyer 2's

equilibrium strategy, her reservation price satisfies  $p_1^* - c_L + \delta(c_H - c_L) = \delta(1 - \alpha)(c_H - c_L)$ , so  $p_1^* = c_L - \delta\alpha(c_H - c_L)$ . Rejection therefore requires  $p_1^* > v_L$ , ensuring that even Buyer 1's highest possible low offer is rejected on path, or equivalently  $\delta \leq \bar{\delta}/\alpha$ . Thus, the equilibrium can be sustained only when the seller is sufficiently impatient that continuation payoffs do not make acceptance of the low offer optimal.

This behavior makes the period-1 outcome informative about Buyer 1's signal. After trade, Buyer 2 infers a favorable signal and is sufficiently optimistic that he does not acquire additional information and instead offers  $c_H$ ; the corresponding condition,  $\mu_2^t > \hat{\mu}$ , is automatically satisfied throughout the region. After no trade, he infers an unfavorable signal but still finds private information valuable, requiring

$$\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) - \kappa > 0.$$

This condition generates the lower bound  $\mu_0 > \mu^\sharp$ , while Buyer 1's acquisition incentive imposes the upper bound  $\mu_0 < \bar{\mu}$ . Hence,  $R_3^P$  requires  $\mu^\sharp < \mu_0 < \bar{\mu}$ : the prior must be high enough for Buyer 2 to acquire after no trade, but not so high that Buyer 1 stops acquiring in period 1; and  $\delta < \bar{\delta}/\alpha$ . The region is non-empty whenever  $\kappa < \kappa_{p,A}$ , where  $\kappa_{p,A}$  is defined in Appendix C.

**Welfare across pure-strategy regions.** In  $R_3^P$ , relative to the opaque benchmark, on-path behavior after no trade in period 1 is unchanged: Buyer 2 acquires his private signal, and the low-quality seller rejects the low offer because her reservation price exceeds  $v_L$ . The welfare difference therefore arises only after trade, when Buyer 2 forgoes private learning and offers  $c_H$  directly. This saves the acquisition cost and increases period-2 trade with both seller types, generating efficient additional trade with the high-quality seller but inefficient additional trade with the low-quality seller.

Expected total surplus under transparency exceeds that under the opaque benchmark

whenever

$$\mu_0 > \frac{(1 - \alpha)[\alpha(c_L - v_L) - \kappa]}{\alpha(1 - \alpha)[(c_L - v_L) + (v_H - c_H)] + \kappa(2\alpha - 1)}.$$

The threshold is the prior at which the extra trade induced by transparency becomes welfare improving. As  $\mu_0$  rises, trade following period 1 trade is increasingly likely to involve the high-quality seller, so the gain from additional efficient trade, together with the saved acquisition cost, eventually outweighs the loss from additional low-quality trade. Since this threshold lies strictly below  $\mu^\sharp$  and  $R_3^P$  requires  $\mu_0 > \mu^\sharp$ , transparency strictly raises welfare throughout the region.

In  $R_2^P$ , transparency leaves expected gains from trade unchanged because Buyer 2 follows the information revealed by the period-1 trade outcome rather than acquiring an independent signal. This changes how trade is distributed across histories, but not its expected frequency with either seller type. The welfare gain therefore comes entirely from savings in information-acquisition costs.

In  $R_1^P$ , Buyer 2 forgoes private learning after observing no trade. This saves information-acquisition costs and reduces inefficient trade with the low-quality seller, but also causes some efficient trade with the high-quality seller to be lost. When  $\alpha < \hat{\alpha}$ , the latter loss dominates and transparency reduces welfare throughout  $R_1^P$ . When  $\alpha > \hat{\alpha}$ , the gains dominate at sufficiently pessimistic priors, but as the prior becomes less pessimistic, the loss from foregone high-quality trade becomes more important and the welfare effect eventually turns negative.

Figure 2 summarizes Buyer 2's history-contingent acquisition behavior and equilibrium welfare across the three pure-strategy regions.

#### 4.1.2 Mixed-strategy equilibria

I now turn to the case in which the low-quality seller is sufficiently patient. Because she places greater weight on her continuation payoff, she may be willing to accept a below-cost offer in period 1 to influence Buyer 2's beliefs. However, greater distortion makes the period-1 trade outcome less informative and increases Buyer 2's incentive to acquire his

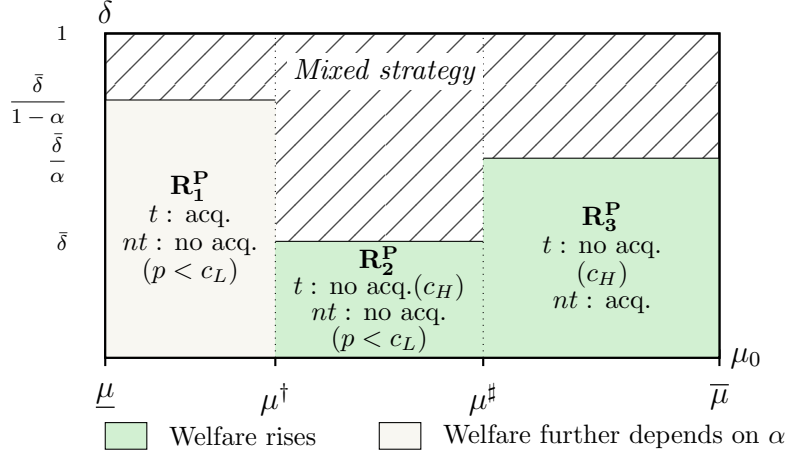


Figure 2: Buyer 2's behavior and the welfare effect of transparency in the pure-strategy equilibrium regions.  $t$  denotes trade and  $nt$  denotes no trade. "Acq." and "no acq." indicate whether Buyer 2 acquires his private signal. Offers made without information acquisition appear in parentheses.

private signal, which in turn weakens the seller's incentive to distort. This feedback rules out a pure-strategy equilibrium and leads to mixing by both players.

On the equilibrium path, the low-quality seller accepts the below-cost offer with positive probability, making the period-1 trade outcome only partially informative about Buyer 1's private signal. Buyer 2 therefore sometimes acquires his private signal after histories in which he would have relied on the trade outcome alone in the pure-strategy equilibrium. Transparency now affects welfare through both the learning effect and the low-quality seller's strategic distortion. Because  $v_L < c_L$ , the resulting period-1 trade is inefficient and reduces social surplus.

**Equilibrium behavior.** The regions of the  $(\mu_0, \delta)$ -space in which mixed-strategy equilibria arise are denoted by  $R_1^M, R_2^M, R_3^M$ , and  $R_4^M$  in Figure 1. Buyer 2's behavior differs across these regions according to the period-1 trade outcome. In each region, he is indifferent about private-signal acquisition after one history and therefore randomizes, while taking a pure acquisition action after the other.

In  $R_1^M$ , after trade, Buyer 2 mixes over private-signal acquisition and offers  $c_H$  if he does

not acquire; after no trade, he does not acquire and instead offers some  $p < c_L$ . In  $R_2^M$ , after trade, he does not acquire and offers  $c_H$ ; after no trade, he mixes over acquisition and offers some  $p < c_L$  if he does not acquire. In  $R_3^M$ , he acquires after trade and mixes over acquisition after no trade, offering some  $p < c_L$  if he does not acquire. In  $R_4^M$ , he mixes over acquisition after trade and offers  $c_H$  if he does not acquire, while after no trade he acquires with certainty. The remaining equilibrium strategies are common across the four regions and are specified in Appendix C.

To illustrate how the mixing conditions sustaining these regions arise, consider  $R_3^M$  as a representative case. Let  $\eta_{R_3^M}$  denote the probability with which the low-quality seller accepts Buyer 1's period-1 offer  $v_L$ , and let  $\beta_{R_3^M}$  denote the probability with which Buyer 2 acquires his private signal after observing no trade. The posteriors  $\mu_2^t$  and  $\mu_2^{nt}$  follow from Bayes' rule and are given in Appendix C.

Because the low-quality seller mixes in period 1, the trade outcome is less informative than in any pure-strategy equilibrium. Observing no trade leads Buyer 2 to revise his belief downward by less, while observing trade leads him to revise it upward by less. The seller's mixing probability is chosen so that, after no trade, Buyer 2 is exactly indifferent between acquiring and not acquiring his private signal:

$$\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) - \kappa = 0.$$

Hence,  $\mu_2^{nt} = \underline{\mu}$ , which pins down  $\eta_{R_3^M} = 1 - \left(\frac{1-\underline{\mu}}{\underline{\mu}}\right) \left(\frac{\mu_0}{1-\mu_0}\right) \left(\frac{1-\alpha}{\alpha}\right)$ , with  $0 < \eta_{R_3^M} < 1$  ensuring interior mixing by the low-quality seller.

Buyer 2's mixing after no trade must, in turn, leave the low-quality seller indifferent between accepting and rejecting  $v_L$ :

$$(v_L - c_L) + \delta(1 - \alpha)(c_H - c_L) = \delta\beta_{R_3^M}(1 - \alpha)(c_H - c_L).$$

This yields  $\beta_{R_3^M} = 1 - \frac{c_L - v_L}{\delta(1 - \alpha)(c_H - c_L)}$ , so interior mixing by Buyer 2 requires  $\delta > \bar{\delta}/(1 - \alpha)$ .

After trade, Buyer 2 strictly prefers to acquire his private signal. If his off-path no-acquisition offer is  $c_H$ , this requires

$$\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) - \kappa > \mu_2^t(v_H - c_H) + (1 - \mu_2^t)(v_L - c_H),$$

together with  $\mu_2^t > \hat{\mu}$ , ensuring that  $c_H$  is optimal conditional on not acquiring. Alternative off-path no-acquisition offers yield outcome-equivalent equilibria.

Combining these conditions,  $R_3^M$  exists for  $\underline{\mu} < \mu_0 < \mu^\dagger$  and  $\delta > \bar{\delta}/(1 - \alpha)$ , with a non-empty belief interval when  $\kappa < \kappa^m$ . Interior mixing additionally requires  $\alpha < 1 - \bar{\delta}/\delta$ . The equilibrium is sustained by mutual indifference: the seller's mixing makes Buyer 2 indifferent after no trade, while Buyer 2's mixing makes the seller indifferent in period 1. As  $\alpha$  rises, Buyer 2 is more likely to identify the low-quality seller and avoid period-2 trade, reducing the continuation value that sustains her mixing.

**Welfare across mixed-strategy regions.** In  $R_3^M$ , transparency lowers welfare because Buyer 2 forgoes private-signal acquisition with positive probability after no trade, causing some efficient trade with the high-quality seller to be lost, while the low-quality seller's mixing also generates inefficient period-1 trade.

In  $R_1^M$ , reduced private-signal acquisition saves information costs, while seller distortion creates inefficient trade. For sufficiently low acquisition costs,  $\kappa < \kappa_{m,A}(\alpha, \delta)$ , the distortion effect dominates and transparency strictly reduces social surplus throughout  $R_1^M$ , as shown in Appendix C.

In  $R_2^M$  and  $R_4^M$ , transparency raises welfare because the gains from reduced information acquisition and high-quality trade outweigh the losses from seller distortion. In  $R_4^M$ , this mechanism is especially direct after period-1 trade, when Buyer 2 forgoes private-signal acquisition while still trading with the high-quality seller.

Together with the pure-strategy analysis, these results establish Theorem 1. Appendix C defines a bound  $\kappa^* > 0$  such that, for every  $\kappa < \kappa^*$ , the otherwise ambiguous welfare effect

in  $R_1^P$  can be signed.

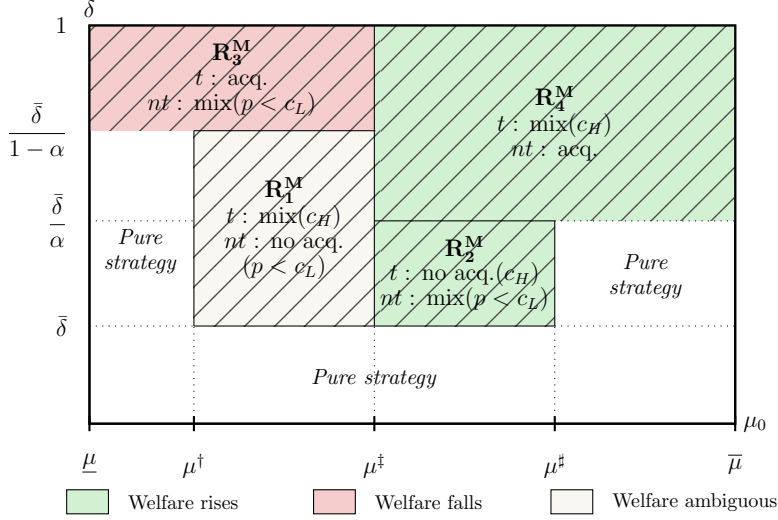


Figure 3: Buyer 2's behavior and the welfare effect of transparency in the mixed-strategy equilibrium regions.  $t$  and  $nt$  denote trade and no trade. Parentheses show offers made without acquisition. Colors indicate the welfare effect of transparency relative to opacity, while hatching identifies mixed-strategy equilibrium regions.

## 4.2 Gains from trade with both types

Consider now the case  $v_L > c_L$ , so trade with the low-quality seller generates positive social surplus. Relative to the case  $v_L < c_L$ , this changes both the informational role of trade and the seller's strategic incentives.

Recall first that, in the opaque benchmark, a buyer who acquires information offers  $c_H$  following a high signal and  $c_L$  following a low signal. Because the low-quality seller accepts the  $c_L$  offer whereas the high-quality seller rejects it, a low signal no longer necessarily leads to no trade. Conditional on information acquisition, no trade can occur only when the seller is high quality and the buyer receives a low signal. Thus, once trade histories become observable, no trade is good news about seller quality, while trade is relatively more likely for the low-quality seller. This changes both Buyer 2's posterior beliefs and his incentive to acquire his private signal.

The form of strategic distortion also reverses. When  $v_L < c_L$ , the low-quality seller may

accept a loss-making offer in order to mimic the high-quality seller, generating inefficient overtrading. When  $v_L > c_L$ , she would ordinarily accept the low offer, so mimicking the high-quality seller instead requires rejecting a profitable trade. Transparency can therefore generate inefficient undertrading or delay.

The welfare pattern changes as well. For pessimistic priors, transparency remains welfare-reducing in the relevant equilibrium regions. For optimistic priors, transparency can still improve welfare, but this gain disappears when the seller is sufficiently patient. Most notably, the case  $v_L > c_L$  gives rise to welfare-neutral equilibria: for sufficiently patient sellers and intermediate or optimistic priors, the welfare gains and losses induced by transparency exactly offset. The analysis below characterizes these differences formally.

Throughout the analysis, I assume

$$\alpha > \frac{2}{1 + \sqrt{1 + 4 \frac{c_H - v_L}{c_H - c_L}}}. \quad (4)$$

Because  $v_L > c_L$ , this condition necessarily requires  $\alpha > \frac{\sqrt{5}-1}{2} \approx 0.618$ . For a given signal precision, the restriction limits the surplus generated by trade with the low-quality seller. Equivalently, it ensures that the low-quality seller's current trading payoff is not too large relative to the reputational benefit of influencing the period-1 outcome, thereby allowing the relevant mixed-strategy equilibria to arise.

A further difference from the case  $v_L < c_L$  is that Buyer 1's information-acquisition decision may now differ across market regimes. Because trade with the low-quality seller is efficient, a low signal can still lead to positive surplus. The terms of this trade differ across the transparent and opaque markets, which changes the value of acquiring information. The following lemma characterizes the resulting acquisition regions.

Recall from (3) that  $\underline{\mu}$  and  $\tilde{\mu}$  denote the lower and upper bounds of the acquisition region in the opaque benchmark. For the equilibrium period-1 reservation price  $p_1^*$  of the low-quality seller, let  $\underline{\mu}^{p_1^*}$  and  $\bar{\mu}^{p_1^*}$  denote the corresponding acquisition thresholds in the transparent

market. Their expressions are provided in (A.4) Appendix C.

**Lemma 4.2** (Buyer 1's information acquisition). *For sufficiently low information-acquisition cost  $\kappa$ , Buyer 1 acquires his private signal in both markets when  $\mu_0 \in (\underline{\mu}, \bar{\mu}^{p_1^*})$ , only in the transparent market when  $\mu_0 \in (\underline{\mu}^{p_1^*}, \underline{\mu})$ , and only in the opaque market when  $\mu_0 \in (\bar{\mu}^{p_1^*}, \tilde{\mu})$ . Outside  $(\underline{\mu}^{p_1^*}, \tilde{\mu})$ , he acquires in neither market.*

I first focus on the cases where the initial beliefs lie in the **common-acquisition region**  $\mu_0 \in (\underline{\mu}, \bar{\mu}^{p_1^*})$ , where Buyer 1 acquires information in both the transparent market and the opaque benchmark. Define  $\tilde{\delta} := (v_L - c_L)/(c_H - c_L)$  and  $\alpha' := (c_H - v_L)/(c_H - c_L)$ , and let  $\mu'$  and  $\mu''$  denote additional belief thresholds satisfying  $\underline{\mu}^{p_1^*} < \underline{\mu} < \mu' < \mu'' < \bar{\mu}^{p_1^*} < \tilde{\mu}$ . Their expressions are given in (A.4) in Appendix C.

For ease of exposition, I refer to priors  $\mu_0 \in (\underline{\mu}, \mu')$  as relatively pessimistic,  $\mu_0 \in (\mu', \mu'')$  as intermediate, and  $\mu_0 \in (\mu'', \bar{\mu}^{p_1^*})$  as relatively optimistic. The low-quality seller's equilibrium period-1 reservation price is  $p_1^* = c_L + \delta(c_H - c_L)$  if  $\mu_0 < \mu''$  and  $\delta < \tilde{\delta}$ ,  $p_1^* = c_L + \delta\alpha(c_H - c_L)$  if  $\mu_0 > \mu''$  and  $\delta < \tilde{\delta}/\alpha$ , and  $p_1^* = v_L$  otherwise. Appendix C provides the formal derivation.

**Theorem 2.** *Suppose  $\mu_0 \in (\underline{\mu}, \bar{\mu}^{p_1^*})$  so that buyers' initial beliefs lie in the common-acquisition region. Then, generically:*

- (I) *On the equilibrium path, Buyer 2's probability of acquiring a private signal is lower in the transparent market than in the opaque benchmark.*
- (II) *There exists  $\kappa^{**} > 0$  such that, for every signal acquisition cost  $\kappa < \kappa^{**}$ , the effect of transparency on social surplus relative to the opaque benchmark is:*
  - (A) *If  $\delta < \tilde{\delta}$ , transparency is welfare-reducing for pessimistic and intermediate priors,  $\mu_0 \in (\underline{\mu}, \mu'')$ , and welfare-improving for optimistic priors,  $\mu_0 \in (\mu'', \bar{\mu}^{p_1^*})$ .*
  - (B) *If  $\tilde{\delta} < \delta < \frac{\tilde{\delta}}{\alpha}$ , transparency is welfare-reducing for pessimistic priors,  $\mu_0 \in (\underline{\mu}, \mu')$ , and welfare-improving for optimistic priors,  $\mu_0 \in (\mu'', \bar{\mu}^{p_1^*})$ .*

(C) If  $\delta > \frac{\tilde{\delta}}{\alpha}$ , transparency is welfare-reducing for pessimistic priors,  $\mu_0 \in (\underline{\mu}, \mu')$ , and has no effect on welfare for intermediate and optimistic priors,  $\mu_0 \in (\mu', \bar{\mu}^{p_1^*})$ .

Part (I) of Theorem 2 reflects the learning effect described above: observing the period-1 outcome reduces Buyer 2's incentive to acquire a private signal. Part (II) shows how the welfare effect depends on the interaction between this learning effect and the low-quality seller's strategic undertrading. For sufficiently small acquisition costs, the sign of the welfare effect varies with the prior and the seller's patience.

When  $\delta < \tilde{\delta}$ , the seller is sufficiently impatient that strategic distortion is absent, so welfare is determined entirely by the learning effect. For pessimistic priors, transparency reduces information-acquisition costs but also generates foregone gains from trade with the high-quality seller; the latter effect dominates, so transparency lowers welfare. For optimistic priors, by contrast, transparency both reduces information-acquisition costs and increases period-2 trade with the high-quality seller. The two components of the learning effect therefore reinforce each other, and transparency raises welfare.

When  $\tilde{\delta} < \delta < \tilde{\delta}/\alpha$ , strategic undertrading arises for sufficiently pessimistic and intermediate priors. For pessimistic priors, the welfare loss from delayed low-quality trade outweighs the savings in information-acquisition costs, so transparency lowers welfare. For sufficiently optimistic priors, strategic distortion disappears and transparency raises welfare through the learning effect. For intermediate priors, the welfare effect is ambiguous.

When  $\delta > \tilde{\delta}/\alpha$ , undertrading arises over a broader range of priors. Transparency remains welfare-reducing for relatively pessimistic priors because the loss from delayed efficient trade dominates the information-acquisition cost savings. For intermediate and optimistic priors, however, equilibrium indifference makes the gains and losses induced by transparency exactly offset, leaving welfare unchanged relative to the opaque benchmark. Figure 4 illustrates these regions.

The pure- and mixed-strategy analyses below characterize these equilibrium responses and trace their implications for the welfare regions in Theorem 2.

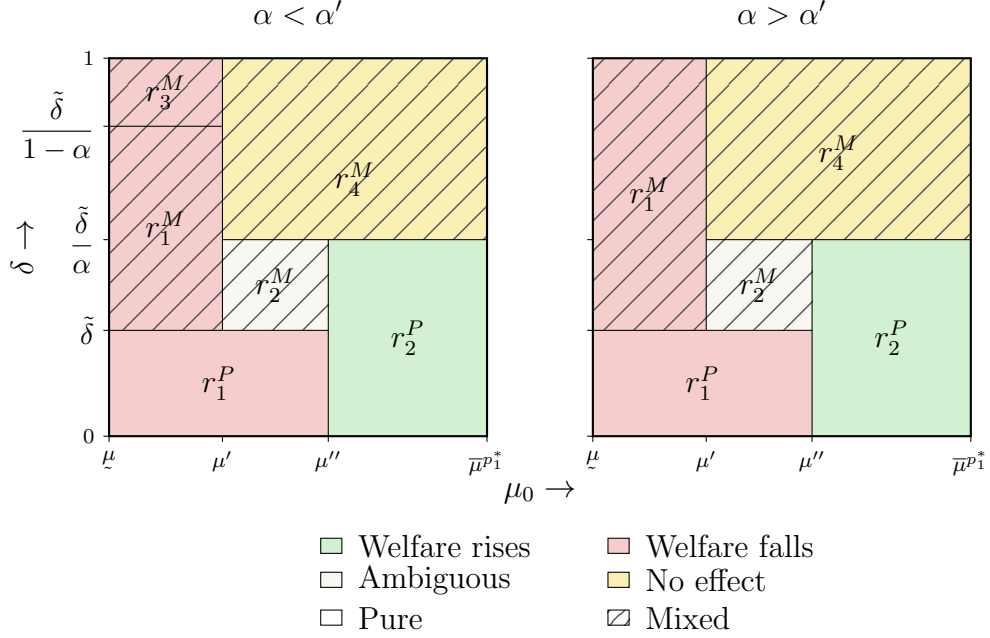


Figure 4: Welfare effect of transparency relative to an opaque market across equilibrium regions.

#### 4.2.1 Pure-strategy equilibria

In a pure-strategy equilibrium, Buyer 1 acquires his private signal and conditions his period-1 offer on its realization. The low-quality seller accepts all offers on-path, whereas the high-quality seller rejects the low offer. Consequently, no trade can occur on the equilibrium path only when Buyer 1 receives a low signal and the seller is high quality. Observing no trade therefore reveals high quality, while observing trade is less informative.

**Equilibrium behavior.** Pure-strategy equilibria arise in two regions,  $r_1^P$  and  $r_2^P$ , of the  $(\mu_0, \delta)$ -space. Their formal definitions are provided in Appendix C, and Figure 4 depicts the corresponding parameter regions. After observing no trade, Buyer 2 does not acquire a private signal and offers  $c_H$ . His behavior after trade differs across the two regions. In  $r_1^P$ , the posterior following trade is sufficiently pessimistic that Buyer 2 does not acquire and offers  $c_L$ . In  $r_2^P$ , the posterior remains sufficiently favorable for private learning to be valuable, so Buyer 2 acquires his signal. The remaining equilibrium strategies are reported

in Appendix C.

**Welfare.** The welfare difference across the two regions is driven by Buyer 2's response to the public history. In  $r_1^P$ , transparency lowers welfare because the loss from foregone efficient trade with the high-quality seller exceeds the savings in information-acquisition costs. In  $r_2^P$ , transparency both reduces information-acquisition costs following a first-period rejection and increases the probability of second-period trade with the high-quality seller. The two effects therefore reinforce one another, so transparency unambiguously raises welfare.

#### 4.2.2 Mixed-strategy equilibria

When the low-quality seller is sufficiently patient, she may be willing to reject an otherwise profitable period-1 offer in order to improve her future reputation. This creates a feedback between her trading behavior and Buyer 2's information-acquisition decision. If the low-quality seller rejects too often, no trade becomes less informative about seller quality, making private information acquisition more attractive to Buyer 2 and reducing the seller's benefit from rejection. This feedback gives rise to mixed-strategy equilibria.

On the equilibrium path, Buyer 1 acquires his private signal and offers  $c_H$  following a high realization and  $v_L$  following a low realization. The low-quality seller accepts the offer  $v_L$  with non-unit probability. Consequently, unlike in the pure-strategy equilibria, observing no trade no longer perfectly reveals high quality. Buyer 2's information-acquisition decision therefore depends on the posterior induced by each trading history.

**Equilibrium behavior.** Mixed-strategy equilibria arise in four regions,  $r_1^M, r_2^M, r_3^M$ , and  $r_4^M$ , of the  $(\mu_0, \delta)$ -space. Their formal definitions and the remaining equilibrium strategies are provided in Appendix C.

In  $r_1^M$ , Buyer 2 does not acquire after observing trade and offers  $c_L$ , while after observing no trade he mixes over information acquisition and offers  $c_H$  if he does not acquire. In  $r_2^M$ , he mixes over acquisition after trade and does not acquire after no trade. In  $r_3^M$ , he again

mixes after trade but acquires with probability one after no trade. Finally, in  $r_4^M$ , he acquires after trade and mixes over acquisition after no trade.

These differences reflect how the two public histories shift Buyer 2's posterior beliefs. For relatively low priors, observing trade provides stronger evidence of low quality, making private learning relatively more valuable after no trade. For relatively high priors, observing no trade shifts beliefs more strongly toward high quality, making private learning relatively less valuable after no trade.

**Welfare.** The welfare effects across the mixed-strategy regions follow from the interaction between reduced information acquisition and strategic rejection. Transparency lowers welfare in  $r_1^M$  and  $r_3^M$ , has an ambiguous effect in  $r_2^M$ , and leaves welfare unchanged in  $r_4^M$ . In the welfare-reducing regions, the losses from delayed efficient trade outweigh the gains from reduced information acquisition. In  $r_4^M$ , equilibrium indifference causes the gains and losses induced by transparency to offset exactly.

The preceding pure- and mixed-strategy analyses together establish Theorem 2. Appendix C defines a bound  $\kappa^{**} > 0$  such that, for every  $\kappa < \kappa^{**}$ , the welfare effects defined in the Theorem 2 hold.

#### 4.2.3 Information acquisition outside the common-acquisition region

When buyers' initial beliefs are outside the common-acquisition region, transparency encourages Buyer 1's acquisition for relatively pessimistic priors but discourages it for more optimistic priors, as established in Lemma 4.2. For pessimistic priors, the higher low-quality seller reservation price makes information acquisition more valuable; for optimistic priors, it lowers the gain from conditioning the offer on a low signal. The welfare comparison therefore reflects the same forces as before, but with Buyer 1's acquisition behavior also differing across markets. The following results characterize the two regions.

**Lemma 4.3** (Acquisition only under transparency). *Suppose that  $\mu_0 \in (\underline{\mu}^{p^*}, \underline{\mu})$ , so Buyer 1*

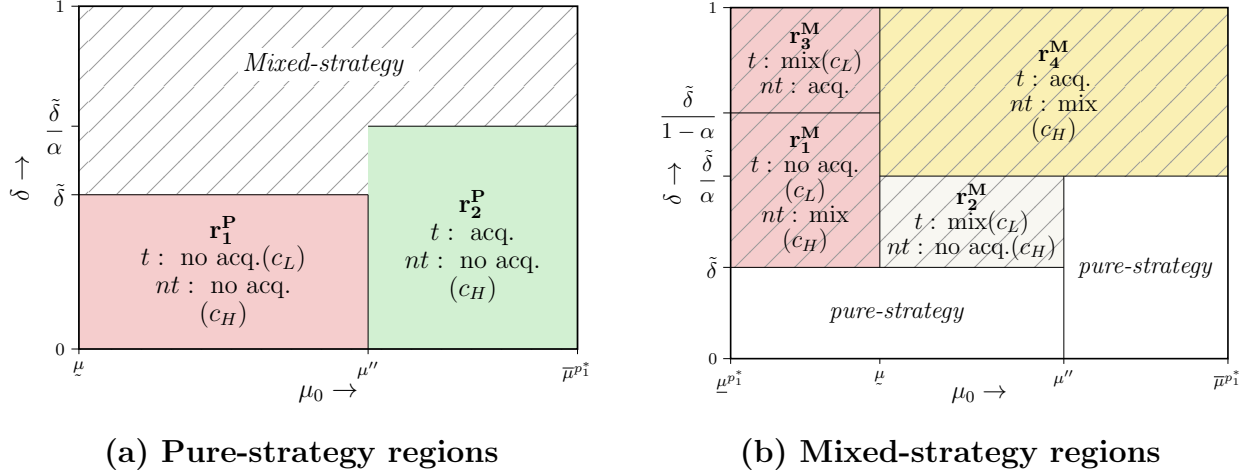


Figure 5: Buyer 2's equilibrium behavior and the welfare effect of transparency across pure- and mixed-strategy equilibrium regions. Panels (a) and (b) show the pure- and mixed-strategy regions, respectively.  $t$  and  $nt$  denote trade and no trade, and parentheses indicate offers made without information acquisition.

*acquires his private signal under transparency but not in the opaque benchmark. Relative to the opaque benchmark, transparency is welfare-improving for sufficiently small  $\kappa$ .*

The proof, including Buyer 2's history-contingent behavior and the associated welfare comparison, is provided in Appendix C. Absent seller distortion, trade is bad news about quality and no trade is good news, so Buyer 2 offers  $c_L$  after trade. In the mixed equilibrium, his response after no trade depends on the extent of seller distortion. The welfare gain arises because transparency enables efficient trade with high-quality sellers that the opaque market would otherwise miss; for sufficiently small acquisition costs, this gain outweighs the cost of information acquisition.

**Lemma 4.4** (Acquisition only in the opaque market). *Suppose that  $\mu_0 \in (\bar{\mu}^{p*}, \tilde{\mu})$ . In this region, Buyer 1 acquires his private signal in the opaque benchmark but not under transparency. Under transparency, Buyer 1 offers  $c_H$  without acquiring his signal, and Buyer 2 acquires his signal following on-path trade. Transparency yields strictly higher welfare than*

*the opaque benchmark.*

When  $\mu_0 \in (\bar{\mu}^{p^*}, \tilde{\mu})$ , transparency can support an equilibrium in which Buyer 1 does not acquire information and offers  $c_H$ , while Buyer 2 acquires information after observing trade. Since trade occurs with probability one on path, Buyer 2's posterior remains at the prior, so his acquisition incentives coincide with those in the opaque benchmark. Off-path beliefs after no trade raise the low-quality seller's period-1 reservation price above  $c_L$ , which makes Buyer 1's non-acquisition optimal over this interval. Relative to opacity, the same trade allocation is then achieved with one fewer private signal acquired, so transparency strictly raises welfare. The proof is provided in Appendix C.

## 5 Discussion

The baseline model is deliberately parsimonious in order to isolate the interaction between market transparency and endogenous information acquisition. In this section, I present additional analyses and extensions of the model.

**Stochastic Buyer Arrival.** In the baseline model, no trade occurs only if the seller rejects the buyer's offer. To introduce search frictions, suppose no buyer arrives in period 1 with probability  $\lambda \in (0, 1)$ , while a buyer arrives in period 2 with probability one. In the transparent market, Buyer 2's posterior after observing  $\tau = nt$  is

$$\mu_2^{nt} = \frac{[\lambda + (1 - \lambda)(1 - \alpha)]\mu_0}{[\lambda + (1 - \lambda)(1 - \alpha)]\mu_0 + q_L(1 - \mu_0)}.$$

Here,  $q_L$  is the total probability of no trade for the low-quality seller. When  $v_L < c_L$ , the seller accepts a loss-making offer with probability  $\eta$ , so  $q_L = \lambda + (1 - \lambda)\alpha(1 - \eta)$ . When  $v_L > c_L$ , the seller rejects a profitable offer with probability  $\eta$ , so  $q_L = \lambda + (1 - \lambda)\alpha\eta$ . In pure-strategy equilibria,  $\eta = 0$ . The posterior following trade is unchanged. Hence, search frictions make no trade less informative, shifting  $\mu_2^{nt}$  toward the prior. For sufficiently small

$\lambda$ , the equilibrium is locally unchanged. Strict acquisition decisions persist by continuity, while under indifference the seller's mixing probability adjusts to  $\eta(\lambda)$ , with  $\eta(\lambda) \rightarrow \eta(0)$  as  $\lambda \rightarrow 0$ . Buyer 2's mixing probability remains unchanged,  $\beta(\lambda) = \beta(0)$ , because the seller's indifference condition is conditional on a buyer arriving and therefore does not depend on  $\lambda$ . Thus, as  $\lambda \rightarrow 0$ , beliefs and  $\eta(\lambda)$  converge to their baseline values, while  $\beta$  is unchanged.

**Comparative statics (private signal accuracy).** A higher private-signal accuracy  $\alpha$  affects welfare through both buyer learning and seller behavior. Holding Buyer 2's continuation strategy fixed, greater accuracy makes the low-quality seller more likely to be correctly identified and therefore directly weakens her incentive to distort period-1 trade. At the same time, however, a higher  $\alpha$  can change Buyer 2's equilibrium acquisition behavior, which alters the seller's continuation payoffs and can strengthen her incentive to distort. The overall welfare effect is therefore ambiguous. The following lemma gives a sufficient condition under which greater signal accuracy reduces welfare when  $v_L < c_L$ .

**Lemma 5.1** (Signal precision and the value of transparency). *Assume  $v_L < c_L$ . For some parameter values and a sufficiently patient seller, holding the prior belief fixed, an increase in signal precision can reverse the welfare ranking of the transparent and opaque markets.*

The proof is given in Appendix C. The mechanism operates through Buyer 2's response to no trade. If Buyer 2 acquires information, the low-quality seller receives the high offer  $c_H$  with probability  $1 - \alpha$ ; if instead he offers  $v_L$  without acquiring, her continuation surplus is zero. Thus, when greater signal precision changes Buyer 2's behavior in this way, it increases the low-quality seller's incentive to induce trade in period 1, strengthening inefficient overtrading. For some parameter values, this indirect effect can dominate the direct reduction in distortion and reverse the welfare ranking of transparency and opacity. Under other parameter configurations, greater signal accuracy may instead improve welfare.

**Noisy observation of trade history.** Consider the  $v_L < c_L$  regime, in which the low-quality seller distorts by accepting the loss-making offer  $v_L$ . Suppose the period-1 outcome is recorded with probability  $\rho$  and lost with probability  $1 - \rho$ . Let  $U_t$  denote the continuation value after an observed trade,  $U_{nt}$  the continuation value after an observed no-trade outcome, and  $U_\emptyset$  the continuation value when the record is lost.

If the seller accepts, her payoff is  $v_L - c_L + \delta[\rho U_t + (1 - \rho)U_\emptyset]$ . If she rejects, her payoff is  $\delta[\rho U_{nt} + (1 - \rho)U_\emptyset]$ . Indifference therefore requires

$$v_L - c_L + \delta[\rho U_t + (1 - \rho)U_\emptyset] = \delta[\rho U_{nt} + (1 - \rho)U_\emptyset].$$

Because the null record occurs with the same probability after either action, the  $U_\emptyset$  terms cancel. Rearranging gives  $c_L - v_L = \delta\rho(U_t - U_{nt})$ .

For the  $v_L > c_L$  case, where the low-quality seller instead distorts by rejecting the profitable offer  $v_L$ , the analogous indifference condition is  $v_L - c_L = \delta\rho(U_{nt} - U_t)$ . Thus, in both cases, imperfect observability scales the intertemporal incentive generated by public history through the product  $\delta\rho$ . As  $\rho$  falls, a larger continuation-value gap is required to sustain seller indifference. Buyer 2's mixing probability  $\beta$  adjusts continuously so that this continuation-value gap is large enough to preserve seller indifference, with the favored history rewarded more strongly as  $\rho$  falls. As  $\rho \rightarrow 1$ , the equilibrium converges to the baseline transparent-market case.

**Comparative statics (seller patience).** As  $\delta$  increases, the low-quality seller's incentive to mimic the high-quality seller increases, strengthening the seller-distortion channel. This exerts downward pressure on the relative welfare of transparency. This effect is strongest under pessimistic priors, where the low-quality seller has more to gain from improving future beliefs and is therefore more willing to distort current trade. For optimistic priors, by contrast, beliefs are already favorable, so the seller has less to gain from manipulating the public trade history and the distortion remains relatively weak. Holding all other parameters

fixed, the expansion of this distortion behavior can be understood by moving vertically upward in the equilibrium figures, though the ultimate net welfare effect also depends on the corresponding discrete shifts in buyers' information acquisition.

**A continuum of mixed-strategy implementations.** The paper focuses on mixed-strategy profiles in which Buyer 1 offers  $v_L$  after a low signal, the low-quality seller accepts with probability  $\eta$ , and Buyer 2 acquires with probability  $\beta$  after the relevant public history. This is without loss for the equilibrium outcomes of interest. More generally, Buyer 1 may offer  $v_L$  with probability  $\lambda \in (0, 1]$  after a low signal and some  $\underline{p} < v_L$  otherwise.

Because the low-quality seller rejects  $\underline{p}$ , trade following a low signal and a low-quality seller occurs with probability  $\lambda\eta$ . Since Buyer 2 observes only the trade outcome, his posterior beliefs and continuation behavior depend on  $\lambda$  and  $\eta$  only through this product. Thus, if  $\eta^*$  is the acceptance probability in the main-text equilibrium, any pair satisfying

$$\lambda\eta = \eta^*$$

induces the same public-history distribution and hence the same beliefs, acquisition probability  $\beta$ , and continuation offers.

These implementations differ only in how the same period-1 trade probability is generated. Because Buyer 1 earns zero current surplus both when  $v_L$  is accepted and when an offer below  $v_L$  is rejected, and the low-quality seller is indifferent between accepting and rejecting  $v_L$ , expected payoffs are unchanged. Hence, the main-text mixed equilibria are part of a continuum of implementations with identical public histories, continuation behavior, allocation probabilities, and expected surplus.

## 6 Conclusion

Market transparency is often viewed as a means of reducing information asymmetries and improving market efficiency. This paper shows that this conclusion can change when information acquisition is costly and sellers strategically respond to the histories that future buyers observe. In a dynamic adverse-selection environment, transparency affects welfare through the interaction of public learning and strategic distortion. Public histories can reduce costly private information acquisition, but they also create incentives for low-quality sellers to manipulate current trade in order to influence future beliefs.

The welfare effect depends on how these forces interact. Transparency can increase welfare when the information conveyed by the public history allows buyers to economize on information acquisition while supporting more efficient trade than in the opaque market. It can instead reduce welfare when buyers rely on a misleading history and forgo private information that would have supported efficient trade.

Seller manipulation is unambiguously welfare-reducing because it distorts the seller's trading decision away from the efficient allocation, either by inducing inefficient trade or by preventing efficient trade. On top of this direct allocative loss, the distortion changes the informativeness of the public history and thereby affects Buyer 2's information-acquisition decision. The form of the trading distortion depends on the efficiency of low-quality trade: transparency can induce overtrading when  $v_L < c_L$  and undertrading when  $v_L > c_L$ . In the latter case, welfare-neutral equilibria can also arise when the gains and losses generated by transparency exactly offset.

The welfare comparison can be highly sensitive to the informational environment. In particular, transparent markets can perform worse as the quality of buyers' private information improves. Holding all other primitives fixed, an arbitrarily small increase in signal precision can induce a discrete change in equilibrium behavior and reverse the welfare ranking between the transparent and opaque markets. The underlying tradeoff also persists when buyer arrival is stochastic and when past trade is observed only noisily. Taken together, the

results qualify the conventional presumption that greater market transparency is necessarily welfare-improving: when public histories are shaped by strategic behavior and substitute for costly private learning, their value depends on both the information they convey and the incentives they create.

## References

- Aghamolla, Cyrus, and Shunsuke Matsuno.** 2026. “Mandatory vs. Voluntary Disclosure in the Dynamic Market for Lemons.” Available at SSRN.
- Aghamolla, Cyrus, and Tadashi Hashimoto.** 2023. “Endogenous Offer Arrival in the Market for Lemons.” Working paper.
- Asriyan, Vladimir, William Fuchs, and Brett Green.** 2017. “Information Spillovers in Asset Markets with Correlated Values.” *American Economic Review*, 107(7): 2007–40.
- Dilmé, Francisc.** 2025. “Repeated Trade with Imperfect Information about Previous Transactions.” *Theoretical Economics*. 209–254.
- Fuchs, William, Aniko Öry, and Andrzej Skrzypacz.** 2016. “Transparency and distressed sales under asymmetric information.” *Theoretical Economics*, 11(3): 1103–1144.
- Horton, John J., and Lydia B. Chilton.** 2010. “The Labor Economics of Paid Crowdsourcing.” *Proceedings of the 11th ACM Conference on Electronic Commerce*, 209–218.
- Hwang, Ilwoo.** 2018. “Dynamic Trading with Developing Adverse Selection.” *Journal of Economic Theory*, 176: 761–802.
- Jang, Inkee, and Kee-Youn Kang.** 2021. “Adverse Selection and Costly Information Acquisition in Asset Markets.” *Journal of Mathematical Economics*, 97: 102533.
- Kaya, Ayça, and Santanu Roy.** 2022a. “Market Screening with Limited Records.” *Games and Economic Behavior*, 132: 106–132.
- Kaya, Ayça, and Kyungmin Kim.** 2018. “Trading Dynamics with Private Buyer Signals in the Market for Lemons.” *Review of Economic Studies*, 85(4): 2318–2352.
- Kaya, Ayça, and Qingmin Liu.** 2015. “Transparency and Price Formation.” *Theoretical Economics*, 10(2): 341–383.
- Kaya, Ayça, and Santanu Roy.** 2022b. “Price Transparency and Market Screening.” *Working Paper*. SSRN Working Paper.
- Kaya, Ayça, and Santanu Roy.** 2024. “Repeated Trading: Transparency and Market Structure.” *American Economic Review*, 114(8): 2388–2435.
- Kim, Kyungmin.** 2017. “Information about Sellers’ Past Behavior in the Market for Lemons.” *Journal of Economic Theory*, 169: 365–399.
- Pallais, Amanda.** 2014. “Inefficient Hiring in Entry-Level Labor Markets.” *American Economic Review*, 104(11): 3565–3599.
- Ravid, Doron, Anne-Katrin Roesler, and Balázs Szentes.** 2022. “Learning before Trading: On the Inefficiency of Ignoring Free Information.” *Journal of Political Economy*, 130(2): 346–387.
- Vairo, Maren, and Piotr Dworczak.** 2023. “Optimal Transparency in OTC Markets.” Available at SSRN.

## A General Result

**Lemma A.1.** *In any PBE (opaque or transparent market), no buyer offers more than  $c_H$ , and the high-quality seller earns zero profit. Her reservation price in both periods is  $c_H$ , and she accepts  $c_H$  whenever it is offered on path.*

*Proof.* Any offer  $p > c_H$  cannot be optimal, since the buyer can lower it slightly while remaining above  $c_H$ , inducing the same acceptance behavior at a lower price. Hence, no buyer offers more than  $c_H$ . The high-quality seller therefore earns zero profit in period 2, so her continuation value after rejecting in period 1 is also zero and her reservation price in both periods is  $c_H$ . Finally, she cannot reject  $c_H$  with positive probability on path: since  $v_H > c_H$ , the buyer could profitably deviate to  $c_H + \varepsilon$  for sufficiently small  $\varepsilon > 0$  and induce acceptance with probability one. Thus, the high-quality seller accepts  $c_H$  on path and earns zero equilibrium profit.  $\square$

**Lemma A.2.** *In any PBE, the low-quality seller accepts an offer of  $c_H$  with certainty.*

*Proof.* Suppose a buyer offers  $p = c_H$  in period 1. By accepting, the low-quality seller earns  $c_H - c_L$  immediately and a nonnegative continuation payoff, so her total payoff is at least  $c_H - c_L$ . By rejecting, she earns nothing in period 1, and since Buyer 2 never offers more than  $c_H$  in period 2 (Lemma A.1), her continuation payoff is at most  $c_H - c_L$ , yielding a total payoff of at most  $\delta(c_H - c_L)$ . Since  $\delta < 1$  and  $c_H > c_L$ , accepting strictly dominates rejecting. Hence, the low-quality seller accepts  $c_H$  with probability one.  $\square$

## B Opaque Market

### Proof of Proposition 1

**Step 1:** The seller accepts iff the offer exceeds her cost as her problem is static.

**Step 2:** The cutoff belief  $\hat{\mu}$  is defined in Equation (1). The buyer offers  $c_H$  iff  $\mu \geq \hat{\mu}$ ; otherwise, she offers  $\underline{p} = c_L$  when  $v_L > c_L$ , and any  $\underline{p} < c_L$  when  $v_L < c_L$ .<sup>1</sup>

**Step 3:** Let  $p_L := \min\{v_L, c_L\}$ .<sup>2</sup> Define

$$\bar{\kappa}(\mu_0, \alpha) := \alpha\mu_0(v_H - c_H) + (1 - \alpha)(1 - \mu_0)(p_L - c_H) - \max\{\mu_0 - \hat{\mu}, 0\}(v_H - p_L). \quad (\text{A.1})$$

<sup>1</sup>In the latter case, the buyer prefers rejection, and any offer below  $c_L$  is rejected.

<sup>2</sup>For the proof I use  $p_L$  instead of  $\underline{p}$ .  $p_L$  generates the same acceptance behavior as  $\underline{p}$  hence results in the same incentive constraint for the buyer, but makes the analysis cleaner notationally by allowing me to unify both cases  $v_L \gtrless c_L$ .

Buyer  $i$  acquires his private signal if and only if  $\kappa < \bar{\kappa}(\mu_0, \alpha)$ , and offers  $c_H$  after  $s^i = h$ , and  $p$  after  $s^i = l$ . Otherwise, he offers  $c_H$  if  $\mu_0 > \hat{\mu}$  and  $p$  if  $\mu_0 < \hat{\mu}$ .

*Proof of Step 3.* It follows from Equation (3) that  $\hat{\mu} = \frac{c_H - p_L}{v_H - p_L}$ . By Step 2, the buyer offers  $c_H$  iff his belief exceeds  $\hat{\mu}$ , and offers  $p_L$  otherwise. Acquisition is valuable only if the signal can change the buyer's optimal offer, i.e.,  $\mu_i^l < \hat{\mu} < \mu_i^h$ .

If  $\mu_0 > \hat{\mu}$ , then without acquiring the buyer offers  $c_H$ , with payoff  $\mu_0(v_H - c_H) + (1 - \mu_0)(v_L - c_H)$ . With acquisition, the buyer offers  $c_H$  after  $s^i = h$  and  $p_L$  after  $s^i = l$ , yielding expected payoff  $\mu_0\alpha(v_H - c_H) + (1 - \mu_0)[\alpha(v_L - p_L) + (1 - \alpha)(v_L - c_H)] - \kappa$ . Comparing the payoffs and rearranging the payoff inequality yields  $\kappa < -(1 - \alpha)\mu_0(v_H - c_H) + \alpha(1 - \mu_0)(c_H - p_L)$ . Since the definition of  $\hat{\mu}$  implies  $\hat{\mu}(v_H - p_L) = c_H - p_L$ , this is equivalent to  $\kappa < \alpha\mu_0(v_H - c_H) + (1 - \alpha)(1 - \mu_0)(p_L - c_H) - (\mu_0 - \hat{\mu})(v_H - p_L) = \bar{\kappa}(\mu_0, \alpha)$ .

If instead  $\mu_0 < \hat{\mu}$ , then without acquiring the buyer offers  $p_L$ , with payoff  $(1 - \mu_0)(v_L - p_L)$ , while with acquisition his expected payoff is as above. Comparing this payoff with the payoff from acquiring information and rearranging yields  $\kappa < \alpha\mu_0(v_H - c_H) + (1 - \alpha)(1 - \mu_0)(p_L - c_H)$ . Since  $\max\{\mu_0 - \hat{\mu}, 0\} = 0$ , this is precisely  $\kappa < \bar{\kappa}(\mu_0, \alpha)$ .  $\square$

Combining the preceding conditions, Buyer  $i$  acquires the signal iff  $\kappa < \bar{\kappa}(\mu_0, \alpha)$ . Since  $\bar{\kappa}(\mu_0, \alpha)$  is single-peaked in  $\mu_0$  and attains its maximum at  $\hat{\mu}$ , this inequality holds for some prior iff  $\kappa$  is below that maximum. Hence, acquisition occurs for a nonempty interval of priors iff  $\kappa < (2\alpha - 1)\hat{\mu}(v_H - c_H)$ , equivalently  $\alpha > \frac{1}{2} + \frac{\kappa}{2\hat{\mu}(v_H - c_H)}$ . Rearranging the acquisition condition  $\kappa < \bar{\kappa}(\mu_0, \alpha)$  gives  $\mu_0 \in (\mu_-^a, \mu_+^a)$ , where  $\mu_-^a = \frac{(1 - \alpha)(c_H - p_L) + \kappa}{(1 - \alpha)(c_H - p_L) + \alpha(v_H - c_H)}$  and  $\mu_+^a = \frac{\alpha(c_H - p_L) - \kappa}{\alpha(c_H - p_L) + (1 - \alpha)(v_H - c_H)}$ , with  $\mu_-^a < \hat{\mu} < \mu_+^a$  whenever the interval is nonempty. It can be verified that  $(\mu_-^a, \mu_+^a)$  coincides with the acquisition region  $\mathcal{A}$  defined in (3), with  $\mu_-^a$  and  $\mu_+^a$  denoting its generic lower and upper bounds. Their case-specific expressions for  $v_L < c_L$  and  $v_L > c_L$  are given in (A.2) and (A.4), respectively.

Otherwise, the buyer does not acquire, his belief remains  $\mu_0$ , and the result follows from Step 2.  $\square$

**Welfare.** When Buyer  $i$  acquires the signal, the following tables summarize social surplus in an opaque market for the case  $v_L < c_L$ . Expected total surplus (ETS) is obtained by weighting each realization by its corresponding probability. A similar table yields the result for  $v_L > c_L$ . Let  $ETS_{\text{opaque}}^a$  denote the ex-ante expected social surplus when Buyer  $i$  acquires the signal:

$$ETS_{\text{opaque}}^a = \begin{cases} (1 + \delta) [\mu_0\alpha(v_H - c_H) + (1 - \mu_0)(v_L - c_L) - \kappa], & \text{if } v_L > c_L, \\ (1 + \delta) [\mu_0\alpha(v_H - c_H) + (1 - \mu_0)(1 - \alpha)(v_L - c_L) - \kappa], & \text{if } v_L < c_L. \end{cases}$$

When Buyer  $i$  does not acquire the signal, his belief at the offer stage remains  $\mu_0$ , so

| $\theta$ | Signal | Probability          | Seller surplus            | Buyer surplus                                  | Total surplus                                  |
|----------|--------|----------------------|---------------------------|------------------------------------------------|------------------------------------------------|
| $H$      | $hh$   | $\alpha^2$           | 0                         | $(1 + \delta)(v_H - c_H) - (1 + \delta)\kappa$ | $(1 + \delta)(v_H - c_H) - (1 + \delta)\kappa$ |
| $H$      | $hl$   | $\alpha(1 - \alpha)$ | 0                         | $v_H - c_H - (1 + \delta)\kappa$               | $v_H - c_H - (1 + \delta)\kappa$               |
| $H$      | $lh$   | $\alpha(1 - \alpha)$ | 0                         | $\delta(v_H - c_H) - (1 + \delta)\kappa$       | $\delta(v_H - c_H) - (1 + \delta)\kappa$       |
| $H$      | $ll$   | $(1 - \alpha)^2$     | 0                         | $-(1 + \delta)\kappa$                          | $-(1 + \delta)\kappa$                          |
| $L$      | $hh$   | $(1 - \alpha)^2$     | $(1 + \delta)(c_H - c_L)$ | $(1 + \delta)(v_L - c_H) - (1 + \delta)\kappa$ | $(1 + \delta)(v_L - c_L) - (1 + \delta)\kappa$ |
| $L$      | $hl$   | $\alpha(1 - \alpha)$ | $c_H - c_L$               | $v_L - c_H - (1 + \delta)\kappa$               | $v_L - c_L - (1 + \delta)\kappa$               |
| $L$      | $lh$   | $\alpha(1 - \alpha)$ | $\delta(c_H - c_L)$       | $\delta(v_L - c_H) - (1 + \delta)\kappa$       | $\delta(v_L - c_L) - (1 + \delta)\kappa$       |
| $L$      | $ll$   | $\alpha^2$           | 0                         | $-(1 + \delta)\kappa$                          | $-(1 + \delta)\kappa$                          |

Table 1: Surplus under signal acquisition in the opaque market when  $v_L < c_L$ .

social surplus follows directly from the equilibrium offer characterized in Step 2. Denote the ex-ante expected social surplus in this case by  $ETS_{\text{opaque}}^n$ :

$$ETS_{\text{opaque}}^n = \begin{cases} (1 + \delta) [\mu_0(v_H - c_H) + (1 - \mu_0)(v_L - c_L)], & \text{if } \mu_0 \geq \hat{\mu}, \\ (1 + \delta) [(1 - \mu_0)(v_L - \min\{v_L, c_L\})], & \text{if } \mu_0 < \hat{\mu}. \end{cases}$$

## C Transparent Market

In this section, I provide the proofs for Theorem 1 and Theorem 2. I begin with a lemma describing player behavior that holds throughout the transparent market case, irrespective of whether trade with the low-quality seller is socially efficient or inefficient.

**Notation.** For Buyer 1, let  $p_1^n$  denote the offer without information acquisition and  $p_1^s$ ,  $s \in \{h, l\}$ , the offer after signal  $s$ . For Buyer 2, let  $p_2^a(\tau)$ ,  $a \in \{n, h, l\}$  and  $\tau \in \{t, nt\}$ , denote his offer after observing period-1 outcome  $\tau$ , where  $a = n$  indicates no acquisition and  $a \in \{h, l\}$  the realized private signal following acquisition. Let  $\mu_i^{\text{int}}$  denote Buyer  $i$ 's belief at the information-acquisition stage and  $\mu_i^s$  his posterior after signal  $s \in \{h, l\}$ . Thus,  $\mu_1^{\text{int}} = \mu_0$ , while  $\mu_2^{\text{int}}$  is his posterior after observing the period-1 outcome and before deciding whether to acquire information. If no information is acquired, the belief remains  $\mu_i^{\text{int}}$ .

**Lemma C.1.** *In the transparent market, Buyer 2 strictly prefers to acquire his private signal iff  $\kappa < \bar{\kappa}(\mu_2^{\text{int}}, \alpha)$ , equivalently iff  $\frac{(1-\alpha)(c_H - \min\{v_L, c_L\}) + \kappa}{(1-\alpha)(c_H - \min\{v_L, c_L\}) + \alpha(v_H - c_H)} < \mu_2^{\text{int}} < \frac{\alpha(c_H - \min\{v_L, c_L\}) - \kappa}{\alpha(c_H - \min\{v_L, c_L\}) + (1-\alpha)(v_H - c_H)}$ , where  $\bar{\kappa}(\cdot, \alpha)$  is defined in (A.1).*

*Proof.* Same as proof of Proposition 1, replacing  $\mu_0$  with  $\mu_2^{\text{int}}$ , since Buyer 2's acquisition decision depends on his interim belief. The seller's period-2 acceptance rule and Buyer 2's offer problem are unchanged.  $\square$

**Equilibrium Selection and Outcome Equivalence:** As discussed in Section 5 of the main text, there exists a continuum of allocation-, payoff-, and welfare-equivalent mixed-strategy equilibria where Buyer 1 randomizes between offering  $v_L$  and a price below  $v_L$  which gets rejected by both types of sellers. To eliminate this payoff- and welfare-irrelevant multiplicity, I impose a strict tie-breaking rule: whenever a buyer is indifferent among offers, he selects the offer that maximizes the probability of trade. Uniqueness claims for mixed-strategy profiles in the following proofs therefore apply up to this outcome equivalence.

## C.1 Gains from trade only with the high-quality seller

This section proves Theorem 1 by establishing the lemmas and propositions on which it depends, in the order in which they appear in the text. In particular, I first prove Lemma 4.1, then characterize the pure- and mixed- strategy equilibria and their welfare implications. Together, these results constitute the proof of Theorem 1.

### C.1.1 Proof of Lemma 4.1

Let  $d_1^T, d_1^O \in \{a, n\}$  denote Buyer 1's information-acquisition decisions in the transparent and opaque markets, respectively. I show that  $d_1^T = d_1^O$  generically.

**Step 1:** Suppose the  $L$ -seller's period-1 reservation price satisfies  $p_1^* \geq v_L$ . Then Buyer 1 cannot earn strictly positive surplus by trading only with the  $L$ -seller, since any accepted offer  $p \geq p_1^*$  yields  $v_L - p \leq 0$ . Thus his best payoff from not making the winning offer  $c_H$  is 0, exactly as in the opaque market where  $v_L < c_L$ . Since the  $H$ -seller's reservation price is  $c_H$  in both markets, Buyer 1 faces the same information-acquisition problem under transparency and opacity, so  $d_1^T = d_1^O$ .

**Step 2:** Now suppose  $p_1^* < v_L$  and consider a pure-strategy equilibrium. If Buyer 1 reaches with positive probability an information set at which an offer below  $c_H$  is optimal,<sup>3</sup> he must induce acceptance by the  $L$ -seller; otherwise he could deviate to some  $p' \in (p_1^*, v_L)$ , which the  $L$ -seller accepts and which yields  $v_L - p' > 0$ . Hence the  $L$ -seller trades whenever Buyer 1 makes an offer below  $c_H$ , and she also trades when he offers  $c_H$ . Therefore  $\Pr(nt \mid L) = 0$ , so on observing no trade Buyer 2 assigns probability one to type  $H$  and offers  $c_H$ . At any accepted offer  $p' < v_L$ , the  $L$ -seller's maximum lifetime payoff from accepting is  $(p' - c_L) + \delta(c_H - c_L)$ , whereas rejecting yields  $\delta(c_H - c_L)$ . Since  $p' < v_L < c_L$ , she strictly prefers rejection, contradicting equilibrium.

If no such information set is reached with positive probability, then even conditional on acquiring the signal, Buyer 1's lowest posterior  $\mu_1^l$  must leave  $c_H$  optimal relative to

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<sup>3</sup>Since the  $H$ -seller's reservation price is  $c_H$ , any accepted offer below  $c_H$  can trade only with the  $L$ -seller.

an offer just above  $p_1^*$ . Hence  $\mu_1^l \geq \hat{\mu}_{p_1^*} := (c_H - p_1^*)/(v_H - p_1^*)$ ,<sup>4</sup> which implies  $\mu_0 \geq \alpha(c_H - p_1^*)/[\alpha(c_H - p_1^*) + (1 - \alpha)(v_H - c_H)] =: \bar{\mu}_{p_1^*}$ . It can be verified that  $\bar{\mu}_{p_1^*} \notin \mathcal{A}((3))$ . Thus, Buyer 1 does not acquire in the opaque market. In transparency, his optimal offer stays same regardless of whether he acquires his signal or not, so information has zero gross value; since  $\kappa > 0$ , he does not acquire there either. Hence  $d_1^T = d_1^O = n$ .

**Step 3:** Finally, consider a mixed-strategy equilibrium in which the  $L$ -seller mixes at an on-path offer  $p_1^*$ . If  $p_1^* < v_L$ , Buyer 1 can deviate to some  $p' \in (p_1^*, v_L)$  sufficiently close to  $p_1^*$ , induce acceptance with probability one, and obtain strictly greater expected surplus. If  $p_1^* > v_L$ , any accepted trade yields  $v_L - p_1^* < 0$ , so Buyer 1 prefers an offer below  $p_1^*$  that is rejected and yields 0. Also, Lemma A.2 establishes that the low-quality seller accepts  $c_H$  with probability one. Hence the only possible on-path mixing price is  $p_1^* = v_L$ , and Step 1 applies.

Combining the cases proves that Buyer 1 faces identical information-acquisition incentives across both regimes. Thus, generically, he acquires his private signal in the transparent market if and only if he acquires it in the opaque market.

### C.1.2 Pure-strategy equilibria

I begin by proving a lemma which is instrumental in characterizing the pure-strategy equilibria.

**Lemma C.2.** *In any pure-strategy PBE in which Buyer 1 acquires his private signal, the period-1 trade outcome perfectly reveals his signal to Buyer 2: trade implies  $s^1 = h$ , whereas no trade implies  $s^1 = l$ .*

*Proof.* Let  $p_1^h$  and  $p_1^l$  denote Buyer 1's high- and low-signal offers. By Step 2 in the proof of Lemma 4.1,  $p_1^* \geq v_L$ . Hence, by Lemma 4.1, Buyer 1 faces the same acquisition and offer problem as in the opaque market. Because acquisition is costly, it is optimal only if the signal changes his action. Thus, as in Step 2 of the proof of Proposition 1,  $\mu_1^l < \hat{\mu} < \mu_1^h$ , so  $p_1^h = c_H > p_1^l$ . I first show that no trade follows  $s^1 = l$ . Suppose  $p_1^* > v_L$ . Any offer  $p_1^l \in [p_1^*, c_H)$  is accepted only by the low-quality seller and yields Buyer 1 a negative payoff, whereas an offer below  $p_1^*$  yields zero. Hence,  $p_1^l < p_1^*$ , and the low-quality seller rejects. Suppose instead that  $p_1^* = v_L$ . If the low-quality seller accepts  $p_1^l = v_L$ , then, because  $p_1^h = c_H$  is accepted by both types, no trade reveals that the seller is high quality. The low-quality seller could then profitably deviate by rejecting  $v_L$ , a contradiction. Therefore, the low-signal offer is rejected in this case as well. Thus, no trade follows  $s^1 = l$ . Since

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<sup>4</sup>Derived analogously to the cutoff in Equation (1).

$p_1^h = c_H$  and both seller types accept  $c_H$ , trade follows  $s^1 = h$ . The period-1 trade outcome therefore perfectly reveals Buyer 1's private signal.  $\square$

**Equilibrium behavior of players in pure-strategy PBE.** Now I construct and characterize the set of pure-strategy equilibria in which Buyer 1 acquires his private signal. Define the following threshold beliefs for the buyers as follows:

$$\begin{aligned}
\underline{\mu} &:= \frac{\kappa + (1 - \alpha)(c_H - v_L)}{\alpha(v_H - c_H) + (1 - \alpha)(c_H - v_L)}, & \mu^\dagger &:= \frac{(1 - \alpha)[\alpha(c_H - v_L) - \kappa]}{\alpha(1 - \alpha)(v_H - v_L) + \kappa(2\alpha - 1)} \\
\mu^\sharp &:= \frac{\alpha[\kappa + (1 - \alpha)(c_H - v_L)]}{\alpha(1 - \alpha)(v_H - v_L) + \kappa(2\alpha - 1)}, & \bar{\mu} &:= \frac{\alpha(c_H - v_L) - \kappa}{\alpha(c_H - v_L) + (1 - \alpha)(v_H - c_H)} \\
\mu^\ddagger &:= \left[ 1 + (1 - \alpha) \frac{\alpha(v_H - c_H) - \kappa}{\kappa + (1 - \alpha)(c_H - v_L)} + \alpha \frac{(1 - \alpha)(v_H - c_H) + \kappa}{\alpha(c_H - v_L) - \kappa} \right]^{-1} \\
\mu^* &:= \frac{\alpha\kappa + \alpha(1 - \alpha)(c_L - v_L)}{(2\alpha - 1)\kappa + \alpha(1 - \alpha)[(v_H - c_H) + (c_L - v_L)]}
\end{aligned} \tag{A.2}$$

The thresholds satisfy  $\underline{\mu} < \mu^\dagger < \hat{\mu} < \mu^\ddagger < \mu^\sharp < \bar{\mu}$  with  $\hat{\mu} = \frac{c_H - v_L}{v_H - v_L}$  (from (1)). Let  $\bar{\delta} = (c_L - v_L)/(c_H - c_L)$ . I begin by describing the equilibrium behavior of the players. This behavior is common across all pure-strategy equilibria.

**Claim 1.** In a pure-strategy equilibrium, the players behave as follows:

1. The high-quality seller accepts an offer  $p$  in either period if and only if  $p \geq c_H$ .
2. The low-quality seller's period-1 reservation cutoff is  $p_1^*$ : she accepts  $p_1 > p_1^*$ , rejects  $p_1 < p_1^*$ , and at equality rejects only when  $p_1^* = v_L$ . In period 2, she accepts iff  $p_2 \geq c_L$ .
3. Buyer 1 acquires his private signal and offers  $p_1^h = c_H$  and  $p_1^l = \tilde{p}$ , where  $\tilde{p}$  is chosen to induce rejection by type  $L$  seller:  $\tilde{p} \leq v_L$  if  $p_1^* = v_L$ , and  $\tilde{p} < p_1^*$  if  $p_1^* > v_L$ . Off path (no acquisition), he offers  $p_1^n = c_H$  iff  $\mu_0 \geq (c_H - v_L)/(v_H - v_L)$ , and  $p_1^n = \tilde{p}$  otherwise.
4. If Buyer 2 does not acquire his signal, then  $p_2^n(t) = c_H$  and  $p_2^n(nt) < c_L$ . If Buyer 2 acquires his signal, then for each  $\tau \in \{t, nt\}$ ,  $p_2^h(\tau) = c_H$  and  $p_2^l(\tau) < c_L$ .

*Proof of Claim 1.* The high-quality seller's behavior follows from Lemma A.1, and the low-quality seller's behavior from the proof of Lemma C.2. Buyer 1's acquisition incentive constraint is  $\underline{\mu} < \mu_0 < \bar{\mu}$ .<sup>5</sup> His signal-contingent offers are optimal by Lemma C.2, Claim 2, and the off-path offer  $p_1^n$  is optimal given  $\mu_0$  by the same cutoff rule. Buyer 2's signal-contingent offers follow from Lemma C.1. It remains to verify his no-acquisition offers.

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<sup>5</sup>The proof is as in Proposition 1: although the low-signal offer may differ, the induced period-1 trade outcome is the same, yielding the same acquisition constraint for Buyer 1.

Since Buyer 1 acquires in equilibrium, the acquisition condition implies  $\mu_1^h > \hat{\mu} > \mu_1^l$ . By Lemma C.2, trade reveals  $s^1 = h$ , so  $\mu_2^t = \mu_1^h > \hat{\mu}$  and the cutoff rule gives  $p_2^n(t) = c_H$ . No trade reveals  $s^1 = l$ , so  $\mu_2^{nt} = \mu_1^l < \hat{\mu}$  and the cutoff rule makes the specified  $p_2^n(nt) < c_L$  optimal.  $\square$

**Claim 2.** In any pure-strategy PBE, Buyer 2 cannot acquire after both possible period-1 trade outcomes. Equivalently,  $(d_2(t), d_2(nt)) \in \{(n, n), (a, n), (n, a)\}$ .

*Proof of Claim 2.* Suppose, to the contrary, that  $d_2(t) = d_2(nt) = a$ . By Lemma C.2, the period-1 trade outcome reveals Buyer 1's signal, so  $\mu_2^t = \mu_1^h$  after trade and  $\mu_2^{nt} = \mu_1^l$  after no trade. If  $\mu_0 \geq \hat{\mu}$ , then Bayes' rule implies  $\mu_2^t = \mu_1^h > \frac{\alpha(c_H - v_L) - \kappa}{\alpha(c_H - v_L) + (1 - \alpha)(v_H - c_H)}$ , so Lemma C.1 implies that acquisition is not optimal after trade, contradicting  $d_2(t) = a$ . If instead  $\mu_0 < \hat{\mu}$ , then  $\mu_2^{nt} = \mu_1^l < \frac{\kappa + (1 - \alpha)(c_H - v_L)}{\alpha(v_H - c_H) + (1 - \alpha)(c_H - v_L)}$ , so Lemma C.1 implies that acquisition is not optimal after no trade, contradicting  $d_2(nt) = a$ . Therefore Buyer 2 cannot acquire at both information sets, leaving only  $(n, n)$ ,  $(a, n)$ , and  $(n, a)$ .  $\square$

Now, I state and derive the equilibrium conditions for each strategy profile corresponding to regions  $R_1^P - R_3^P$  described in Figure 2. Each case corresponds to a full strategy profile in which the behavior of all players (other than acquisition behavior of Buyer 2) is as characterized in Claim 1.

In each case, Buyer 1's incentive constraints for signal acquisition and the off-path non-acquisition offer must also be satisfied. I do not state these constraints separately; instead, they are included in the final conditions for each case. Any alternatively sustainable off-path actions of the players are noted at the end of the proof of each case. Denote the belief of Buyer 2 after observing the period-1 trade outcome  $\{t, nt\}$  as  $\mu_2^t / \mu_2^{nt}$ . From Lemma C.2 (and Bayes' rule) it follows that  $\mu_2^t = \frac{\alpha\mu^0}{\alpha\mu^0 + (1 - \alpha)(1 - \mu^0)}$  ;  $\mu_2^{nt} = \frac{(1 - \alpha)\mu^0}{(1 - \alpha)\mu^0 + \alpha(1 - \mu^0)}$ .

A. *Buyer 2 acquires after trade and does not acquire after no trade.* (corresponds to region  $R_1^P$ )

The low-quality seller's reservation price is  $p_1^* = c_L - \delta(1 - \alpha)(c_H - c_L)$ , by her indifference condition. Since  $p_1^* \geq v_L$  (Lemma C.2, Claim 1), it follows that  $\delta \leq \frac{c_L - v_L}{(1 - \alpha)(c_H - c_L)}$ .

Buyer 2's acquisition constraints are:

$$(\text{obs. trade}) \quad \alpha\mu^t(v_H - c_H) + (1 - \alpha)(1 - \mu^t)(v_L - c_H) - \kappa > \mu^t(v_H - c_H) + (1 - \mu^t)(v_L - c_H),$$

$$(\text{obs. no trade}) \quad \alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) - \kappa < 0.$$

Since he offers  $p < c_L$  after no trade,  $\mu_2^{nt} < \frac{c_H - v_L}{v_H - v_L}$ ; moreover, his off-path offer after trade must be  $p_2^n(t) = c_H$  (Lemma C.2). Combining these conditions with Buyer 1's constraints, this profile is sustained iff  $\underline{\mu} < \mu_0 < \mu^+$  and  $\delta \leq \frac{\bar{\delta}}{1 - \alpha}$ . This case requires  $p_1^n = \tilde{p}$ : if  $p_1^n = c_H$ , Buyer 1's off-path constraint is incompatible with Buyer 2's no-trade acquisition constraint. It can also be verified that the equilibrium region is non-empty for  $\kappa < \kappa_{p,A} = \frac{\sqrt{\alpha^2(1 - \alpha)^2(v_H - v_L)^2 + \alpha(1 - \alpha)(2\alpha - 1)^2(c_H - v_L)(v_H - c_H) - \alpha(1 - \alpha)(v_H - v_L)}}{2\alpha - 1}$ .

B. *Buyer 2 doesn't acquire on observing either period-1 trade outcome:* (corresponds to region  $R_2^P$ )

The proof is analogous to Case A. The low-quality seller's reservation price is  $p_1^* = c_L - \delta(c_H - c_L)$ . Since  $p_1^* \geq v_L$  in equilibrium, it follows that  $\delta \leq \frac{c_L - v_L}{c_H - c_L}$ . For Buyer 2, only the post-trade acquisition constraint changes: because he does not acquire, the corresponding acquisition inequality reverses. Moreover, since his no-acquisition offer after trade is  $c_H$ , the offer cutoff requires  $\mu_2^t > \frac{c_H - v_L}{v_H - v_L}$ . All remaining conditions are unchanged. Thus, this case is sustained iff  $\frac{c_H - v_L}{v_H - v_L} < \mu_0 < \min\{\mu^\sharp, \bar{\mu}\}$  and  $\delta \leq \frac{c_L - v_L}{c_H - c_L}$ . Here Buyer 1's off-path constraint under  $p_1^n = c_H$  determines the lower bound.

An outcome-equivalent profile arises when  $p_1^n = \tilde{p}$ . Then only Buyer 1's off-path constraint changes to  $\mu_0 < \frac{c_H - v_L}{v_H - v_L}$ , while all other conditions are unchanged. The resulting bounds are  $\max\{\underline{\mu}, \mu^\dagger\} < \mu_0 < \frac{c_H - v_L}{v_H - v_L}$  and  $\delta \leq \frac{c_L - v_L}{c_H - c_L}$ . Since the two profiles are outcome-equivalent, their equilibrium regions combine to  $\max\{\underline{\mu}, \mu^\dagger\} < \mu_0 < \min\{\mu^\sharp, \bar{\mu}\}$  and  $\delta \leq \bar{\delta}$ . When  $\kappa < \kappa_{p,A}$  then the binding bounds are  $\mu^\dagger < \mu_0 < \mu^\sharp$  and  $\delta \leq \bar{\delta}$ .

C. *Buyer 2 acquires on observing no trade and doesn't acquire on observing trade:* (corresponds to region  $R_3^P$ )

The proof is also analogous to Case A. The low-quality seller's reservation price is  $p_1^* = c_L - \delta\alpha(c_H - c_L)$ . Since  $p_1^* \geq v_L$  in equilibrium,  $\delta < \frac{c_L - v_L}{\alpha(c_H - c_L)}$ . Here Buyer 2 does not acquire after trade and acquires after no trade, so the corresponding acquisition inequalities from Case A reverse. Since his no-acquisition offer after trade is  $c_H$ , the offer cutoff requires  $\mu_2^t > \frac{c_H - v_L}{v_H - v_L}$ . Analogously, his no-acquisition offer after no trade must satisfy  $p_2^n(nt) < c_L$ . Thus, this case is sustained iff  $\mu^\sharp < \mu_0 < \bar{\mu}$  and  $\delta \leq \frac{\bar{\delta}}{\alpha}$ . This profile requires  $p_1^n = c_H$ ; if  $p_1^n = \tilde{p}$ , Buyer 1's off-path constraint is incompatible with Buyer 2's acquisition constraint post observing no trade. Even here, the equilibrium region is non-empty for  $\kappa < \kappa_{p,A}$ .

**Welfare.** The corresponding welfare effects follow from the equilibrium behavior of the players. Computing expected total surplus as in the opaque market yields ( $ETS_A$ ,  $ETS_B$  and  $ETS_C$  refer to the ex-ante expected total surplus in cases A, B and C respectively):

$$\begin{aligned} ETS_A &= \mu_0 \left[ (\alpha + \alpha^2 \delta)(v_H - c_H) - (1 + \alpha \delta) \kappa \right] \\ &\quad + (1 - \mu_0) \left[ (1 - \alpha)(1 + \delta(1 - \alpha))(v_L - c_L) - (1 + \delta(1 - \alpha)) \kappa \right], \\ ETS_B &= (1 + \delta) \left[ \mu_0 \alpha (v_H - c_H) + (1 - \mu_0)(1 - \alpha)(v_L - c_L) \right] - \kappa, \end{aligned}$$

$$ETS_C = \mu_0 \left[ \alpha(1 + (2 - \alpha)\delta)(v_H - c_H) - (1 + (1 - \alpha)\delta)\kappa \right] \\ + (1 - \mu_0) \left[ (1 - \alpha)(1 + \delta + \alpha\delta)(v_L - c_L) - (1 + \alpha\delta)\kappa \right].$$

Compare each of these expressions with the opaque-market surplus. In Case B, Buyer 2 follows the information revealed by Buyer 1's signal rather than acquiring his own. The expected number of trades is therefore unchanged relative to the opaque market:  $2\alpha$  when  $\theta = H$  and  $2(1 - \alpha)$  when  $\theta = L$ . Hence, the welfare gain comes entirely from the saved acquisition cost, so  $ETS_B - ETS_{\text{opaque}} = \delta\kappa > 0$ . In Case C,  $ETS_C > ETS_{\text{opaque}}$  iff  $\mu_0 > \frac{(1-\alpha)[\alpha(c_L - v_L) - \kappa]}{\alpha(1-\alpha)[(c_L - v_L) + (v_H - c_H)] + \kappa(2\alpha - 1)}$ . Since this threshold is strictly below  $\mu^\dagger$ , the condition holds whenever Case C is sustained.

In Case A, so that  $ETS_A > ETS_{\text{opaque}} \iff \mu_0 < \mu^*$  (defined in (A.4)). The welfare ranking therefore depends on the location of  $\mu^*$  relative to the equilibrium interval  $(\underline{\mu}, \mu^\dagger)$ . If  $\alpha < \hat{\alpha}$  and  $0 < \kappa < \min\{\kappa_{p,A}, \kappa^{\text{mono}}\}$ , then  $\mu^* < \underline{\mu}$ , so transparency lowers welfare throughout the equilibrium region.<sup>6</sup> If  $\alpha > \hat{\alpha}$ , then  $\mu^* > \underline{\mu}$  for every feasible  $\kappa$ . For  $\kappa < \kappa_W^*$ ,  $\underline{\mu} < \mu^* < \mu^\dagger$ , so transparency raises welfare for  $\underline{\mu} < \mu_0 < \mu^*$  and lowers welfare for  $\mu^* < \mu_0 < \mu^\dagger$ . For  $\kappa_W^* < \kappa < \kappa_{p,A}$ ,  $\mu^* > \mu^\dagger$ , so transparency raises welfare throughout the equilibrium region.<sup>7</sup>

To maintain a common small- $\kappa$  restriction, define

$$\kappa^{\text{welf}}(\alpha) \equiv \begin{cases} \min\{\kappa_{p,A}, \kappa^{\text{mono}}\}, & \alpha < \hat{\alpha}, \\ \kappa_W^*, & \alpha \geq \hat{\alpha}. \end{cases} \quad (\text{A.3})$$

Under  $0 < \kappa < \kappa^{\text{welf}}(\alpha)$ , the welfare ranking simplifies as follows. If  $\alpha < \hat{\alpha}$ , transparency lowers welfare throughout Case A. If  $\alpha > \hat{\alpha}$ , transparency raises welfare for  $\underline{\mu} < \mu_0 < \mu^*$  and lowers welfare for  $\mu^* < \mu_0 < \mu^\dagger$ .

### C.1.3 Mixed-strategy equilibria

I prove several results that help characterize the mixed-strategy equilibria.

**Lemma C.3.** *In any mixed-strategy equilibrium, the low-quality seller's period-1 reservation price is  $p_1^* = v_L$ .*

*Proof.* Proved in Step 3 of the proof of Lemma 4.1. □

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<sup>6</sup>Here  $\kappa^{\text{mono}} = \frac{(1-\alpha)[(1-\alpha)(c_H - v_L) - \alpha(c_L - v_L)]}{2\alpha - 1}$ . The thresholds  $\kappa^{\text{mono}}$  and  $\kappa_{p,A}$  are not generally ordered: if  $\kappa^{\text{mono}} < \kappa_{p,A}$ , the welfare cutoff crosses the lower equilibrium bound before Case A disappears, allowing a non-monotonic welfare ranking in  $\mu_0$ ; otherwise, the equilibrium disappears first.

<sup>7</sup> $\kappa_W^* = \frac{\alpha(1-\alpha)}{2(2\alpha-1)} \left[ \sqrt{Q^2 + 4(2\alpha-1)(v_H - c_H)(c_H - c_L)} - Q \right]$ , where  $Q = (v_H - c_H) + (1-\alpha)(c_H - v_L) + \alpha(c_L - v_L)$ . Moreover,  $\kappa_W^* < \kappa_{p,A}$  for  $\alpha > \hat{\alpha}$ .

**Equilibrium behavior of players** I now characterize the set of mixed-strategy equilibrium profiles. All the thresholds are defined in (A.2).

**Lemma C.4.** *Generically in  $(\mu_0, \delta)$ , every mixed-strategy PBE has the structure described in Section 4.1; no other randomization pattern can be sustained.*

*Proof. Step 1.* Generically, Buyer 1 does not randomize over acquisition or offers.

*Proof of Step 1.* Whether type  $L$  accepts purely or mixes, her period-1 reservation price satisfies  $p_1^* \geq v_L$ : under pure acceptance this follows from the deviation argument in Claim 1 of Lemma C.2, while under mixing Lemma C.3 gives  $p_1^* = v_L$ . Hence, inducing acceptance by type  $L$  yields Buyer 1 at most zero surplus, while a rejected offer also yields zero. His acquisition incentives therefore coincide with those in the pure-strategy analysis, so he can be indifferent between acquiring and not acquiring only at  $\mu_0 = \underline{\mu}$  or  $\mu_0 = \bar{\mu}$ .

Any offer randomization must be between  $c_H$  and a rejected offer below  $p_1^*$  and therefore requires the offer-stage belief to equal  $\hat{\mu}$ . If Buyer 1 acquires, this requires  $\mu_1^{s^1} = \hat{\mu}$ , which by (A.2) can occur only when  $\mu_0 < \underline{\mu}$  or  $\mu_0 > \bar{\mu}$ , where acquisition is not optimal. If he does not acquire, randomization requires  $\mu_0 = \hat{\mu}$ , where acquisition is strictly optimal. Thus, outside the cutoff values, Buyer 1 cannot randomize over acquisition or offers.  $\square$

**Step 2.** Buyer 2 may randomize over acquisition, but not over payoff-relevant offers.

*Proof of Step 2.* Buyer 2 faces the same offer problem as in the opaque market, with  $\mu_2^{\text{int}}$  replacing  $\mu_0$ . Randomization between  $c_H$  and an offer below  $c_L$  requires his offer-stage belief to equal  $\hat{\mu}$ . If he acquires,  $\mu_2^{s^2} = \hat{\mu}$  can hold only when acquisition is not optimal; if he does not acquire,  $\mu_2^{\text{int}} = \hat{\mu}$  implies that acquisition is optimal. Both conclusions follow from Lemma C.1. Hence, Buyer 2 cannot randomize over offers and may randomize only over acquisition.  $\square$

**Step 3.** Generically, Buyer 2 randomizes if and only if the low-quality seller randomizes.

*Proof of Step 3.* Suppose the  $L$ -seller mixes while Buyer 2 uses a pure acquisition strategy. Her indifference condition is  $v_L - c_L + \delta U_t = \delta U_{nt}$ . By Lemma C.1, Buyer 2's signal-contingent offers are fixed. Evaluating this condition under the four possible acquisition profiles shows that it can hold only at  $\delta \in \{\bar{\delta}, \bar{\delta}/\alpha, \bar{\delta}/(1 - \alpha)\}$ . Hence, if Buyer 2 is pure, type  $L$  can mix only at boundary values of  $\delta$ .

Conversely, suppose Buyer 2 mixes over acquisition while type  $L$  is pure. The period-1 outcome then reveals Buyer 1's signal, so  $\mu_2^t = \mu_1^h$  and  $\mu_2^{nt} = \mu_1^l$ . Buyer 2 can be indifferent only if one of these interim beliefs equals an acquisition cutoff,  $\underline{\mu}$  or  $\bar{\mu}$ . Solving the resulting equalities for  $\mu_0$  either places Buyer 1 outside his acquisition region or yields a regime boundary.

Thus, away from boundary values of  $\mu_0$  and  $\delta$ , Buyer 2 and type  $L$  must either both randomize or both play pure strategies.  $\square$

**Step 4.** Except at boundary priors, Buyer 2 cannot randomize over acquisition after both trade and no trade.

*Proof of Step 4.* Suppose Buyer 2 mixes after both histories, with acquisition probabilities  $\beta, \beta' \in (0, 1)$ . By Step 3, type  $L$  also mixes, accepting  $v_L$  with probability  $\eta \in (0, 1)$ . Indifference at both histories requires  $\mu_2^t, \mu_2^{nt} \in \{\underline{\mu}, \bar{\mu}\}$ .

- (i) If  $\mu_2^t = \mu_2^{nt} = \underline{\mu}$  for , then  $\eta = 2 - \frac{1}{\alpha}$  and  $\mu_0 = \underline{\mu}$ .
- (ii) If  $\mu_2^t = \mu_2^{nt} = \bar{\mu}$  for , then  $\eta = 2 - \frac{1}{\alpha}$  and  $\mu_0 = \bar{\mu}$ .
- (iii) If  $\mu_2^t = \bar{\mu}$  and  $\mu_2^{nt} = \underline{\mu}$ , then  $\mu_0 = \mu^\dagger$ .
- (iv) If  $\mu_2^t = \underline{\mu}$  and  $\mu_2^{nt} = \bar{\mu}$ , then  $\delta(c_H - c_L)[\alpha\beta' + (1 - \alpha)\beta - 1] = c_L - v_L$ , which is impossible because  $\alpha\beta' + (1 - \alpha)\beta - 1 < 0$ .

Thus, the first three assignments arise only at boundary priors, and the third is impossible. Hence Buyer 2 cannot mix after both histories away from these boundaries.  $\square$

By Steps 1–4, Buyer 2 may randomize only over acquisition, aside from outcome-equivalent mixing among rejected offers below  $c_L$ . Such randomization must coincide with mixing by type  $L$  and occur after at most one period-1 outcome. Therefore, only the four configurations described in the mixed-strategy equilibrium in Section 4.1 can arise.  $\square$

Lemma C.4 proves the structure of Buyer 2's acquisition behavior in Figure 3. The following claim lists the candidate offers by Buyer 2 on not acquiring his signal.

**Claim 1.** Let  $\underline{p} < c_L$ . For regions  $R_1^M - R_4^M$  in Figure 3, the candidate non-acquisition offers  $(p_2^n(t), p_2^n(nt))$  that may be consistent with equilibrium are: (A)  $(c_H, \underline{p})$  or  $(\underline{p}, \underline{p})$ ; (B)  $(c_H, \underline{p})$  or  $(c_H, c_H)$ ; (C)  $p_2^n(nt) = \underline{p}$ ; (D)  $p_2^n(t) = c_H$ .

*Proof of Claim 1.* By the offer cutoff rule, Buyer 2's non-acquisition offer is either  $c_H$  or a rejected offer  $\underline{p} < c_L$ . Under the acquisition behavior in each case, only the listed combinations are consistent with type  $L$  mixing; every other combination makes one of her period-1 actions strictly preferable.  $\square$

**Claim 2.** The following behavior is common to all mixed-strategy PBEs:

1. Type  $H$  accepts in either period if and only if  $p \geq c_H$ .
2. Type  $L$  mixes between accepting and rejecting  $v_L$  in period 1 and accepts in period 2 if and only if  $p_2 \geq c_L$ .
3. Buyer 1 acquires his private signal and offers  $p_1^h = c_H$  and  $p_1^l = v_L$ .
4. If Buyer 2 acquires, then  $p_2^h(\tau) = c_H$  and  $p_2^l(\tau) < c_L$  for each  $\tau \in \{t, nt\}$ . His non-acquisition offers are mentioned in the paper body.

*Proof of Claim 2.* The sellers' strategies follow from Lemmas A.1 and C.3. Buyer 1 acquires whenever  $\underline{\mu} < \mu_0 < \bar{\mu}$ .<sup>8</sup> Optimal acquisition requires inducing trade with type  $H$

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<sup>8</sup>See Claim 1 of proof of pure-strategy equilibrium.

after  $s^1 = h$ , so  $p_1^h = c_H$ . Since type  $L$  can mix only at  $v_L$ , the tie-breaking convention implies  $p_1^l = v_L$ . Off path, if Buyer 1 does not acquire, he offers  $p_1^n = c_H$  if and only if  $\mu_0 > \frac{c_H - v_L}{v_H - v_L}$  and offers  $v_L$  otherwise. Buyer 2's signal-contingent offers follow from Lemma C.1; his no-acquisition offers are verified case by case.  $\square$

I now derive the equilibrium conditions for regions  $R_1^M - R_4^M$ . Strategies are as in Claim 2, except for Buyer 2's acquisition behavior, with mixing probability denoted by  $\beta$ . Buyer 1's acquisition and off-path optimality conditions are imposed throughout and incorporated into the final parameter restrictions. If the  $L$ -seller accepts  $v_L$  with probability  $\eta$ , Buyer 2's interim beliefs are  $\mu_2^t = \frac{\alpha\mu_0}{\alpha\mu_0 + (1-\alpha+\alpha\eta)(1-\mu_0)}$  and  $\mu_2^{nt} = \frac{(1-\alpha)\mu_0}{(1-\alpha)\mu_0 + \alpha(1-\eta)(1-\mu_0)}$ .

A. *Buyer 2 mixes over acquisition after trade and does not acquire after no trade.* (corresponds to region  $R_1^M$ )

**Case 1:**  $p_2^n(t) = c_H$  and  $p_2^n(nt) < c_L$ .

Buyer 2's indifference after trade requires  $\alpha\mu_2^t(v_H - c_H) + (1-\alpha)(1-\mu_2^t)(v_L - c_H) - \kappa = \mu_2^t(v_H - c_H) + (1-\mu_2^t)(v_L - c_H)$ , and hence  $\mu_2^t = \bar{\mu}$ . Therefore,  $\eta_A = \left(\frac{1-\bar{\mu}}{\bar{\mu}}\right) \left(\frac{\mu_0}{1-\mu_0}\right) - \frac{1-\alpha}{\alpha}$ , with  $0 < \eta_A < 1$ .

After no trade, no acquisition followed by a rejected offer requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1-\alpha)(1-\mu_2^{nt})(v_L - c_H) - \kappa < 0$  and  $\mu_2^{nt} < \hat{\mu}$ .<sup>9</sup>

Type  $L$ 's indifference at  $v_L$  gives  $(v_L - c_L) + \delta[\beta_A(1-\alpha) + (1-\beta_A)](c_H - c_L) = 0$ , so  $\beta_A = \frac{1}{\alpha} \left(1 - \frac{c_L - v_L}{\delta(c_H - c_L)}\right)$ , with  $0 < \beta_A < 1$  if and only if  $\bar{\delta} < \delta < \frac{\bar{\delta}}{1-\alpha}$ .

If the off-path offer is  $p_1^n = v_L$ , the remaining conditions imply  $\mu^\dagger < \mu_0 < \hat{\mu}$ ; in this profile, Buyer 1's off-path constraint binds. If instead  $p_1^n = c_H$ , only that constraint changes, yielding  $\hat{\mu} < \mu_0 < \mu^\dagger$ . All other equilibrium conditions are identical. Thus, combining the two outcome-equivalent profiles, Case 1 exists if and only if  $\mu^\dagger < \mu_0 < \mu^\ddagger$  and  $\bar{\delta} < \delta < \frac{\bar{\delta}}{1-\alpha}$ .

**Case 2:**  $p_2^n(t) < c_L$  and  $p_2^n(nt) < c_L$ .

This case cannot be sustained because Buyer 2's no-trade acquisition constraint is incompatible with Buyer 1's acquisition constraint.

B. *Buyer 2 mixes over acquisition after no trade and does not acquire after trade.* (corresponds to region  $R_2^M$ )

**Case 1:**  $p_2^n(t) = c_H$  and  $p_2^n(nt) < c_L$ .

Buyer 2's indifference after no trade requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1-\alpha)(1-\mu_2^{nt})(v_L - c_H) - \kappa = 0$ , so  $\mu_2^{nt} = \underline{\mu}$  and  $\eta_B = 1 - \left(\frac{\mu_0}{1-\mu_0}\right) \left(\frac{1-\underline{\mu}}{\underline{\mu}}\right) \left(\frac{1-\alpha}{\alpha}\right)$ , with  $0 < \eta_B < 1$ .

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<sup>9</sup>Recall from Equation (1) that  $\hat{\mu}$  is the belief threshold for offering  $c_H$  rather than a rejected offer  $p < c_L$ .

After trade, no acquisition followed by an offer of  $c_H$  requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) - \kappa < \mu_2^t(v_H - c_H) + (1 - \mu_2^t)(v_L - c_H)$  and  $\mu_2^t > \frac{c_H - v_L}{v_H - v_L}$ .

Type  $L$ 's indifference at  $v_L$  implies  $(v_L - c_L) + \delta(c_H - c_L) = \delta\beta_B(1 - \alpha)(c_H - c_L)$ , so  $\beta_B = \frac{1}{1 - \alpha} \left(1 - \frac{c_L - v_L}{\delta(c_H - c_L)}\right)$ , with  $0 < \beta_B < 1$  if and only if  $\bar{\delta} < \delta < \frac{\bar{\delta}}{\alpha}$ .

Combining these conditions with Buyer 1's constraints yields  $\mu^\dagger < \mu_0 < \mu^\#$ . This case requires  $p_1^n = c_H$ , since  $p_1^n = v_L$  would require  $\mu_0 < \frac{c_H - v_L}{v_H - v_L}$ , contradicting Buyer 2's acquisition constraint. Hence, Case 1 exists if and only if  $\mu^\dagger < \mu_0 < \mu^\#$  and  $\bar{\delta} < \delta < \frac{\bar{\delta}}{\alpha}$ .

**Case 2:**  $p_2^n(t) = c_H$  and  $p_2^n(nt) = c_H$ .

This case cannot be sustained because Buyer 2's relevant acquisition constraint is incompatible with Buyer 1's acquisition constraint.

C. *Buyer 2 mixes over acquisition after no trade and acquires after trade.* (corresponds to region  $R_3^M$ )

Buyer 2's off-path no-acquisition offer after trade may be either  $c_H$  or some  $p < c_L$ ; the resulting profiles are outcome-equivalent. First suppose  $p_2^n(t) = c_H$ .

Buyer 2's indifference after no trade requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) - \kappa = 0$ , so  $\mu_2^{nt} = \underline{\mu}$  and  $\eta_C = 1 - \left(\frac{1 - \underline{\mu}}{\underline{\mu}}\right) \left(\frac{\mu_0}{1 - \mu_0}\right) \left(\frac{1 - \alpha}{\alpha}\right)$ , with  $0 < \eta_C < 1$ .

After trade, acquisition must be optimal, so  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) - \kappa > \mu_2^t(v_H - c_H) + (1 - \mu_2^t)(v_L - c_H)$ , while  $p_2^n(t) = c_H$  requires  $\mu_2^t > \hat{\mu}$ .

Type  $L$ 's indifference at  $v_L$  gives  $(v_L - c_L) + \delta(1 - \alpha)(c_H - c_L) = \delta\beta_C(1 - \alpha)(c_H - c_L)$ , so  $\beta_C = 1 - \frac{c_L - v_L}{\delta(1 - \alpha)(c_H - c_L)}$ , with  $0 < \beta_C < 1$  if and only if  $\delta > \frac{\bar{\delta}}{1 - \alpha}$ .

If Buyer 1's off-path offer is  $p_1^n = v_L$ , the remaining conditions yield

$\left[1 + \alpha \frac{v_H - c_H}{c_H - v_L} + (1 - \alpha) \frac{\alpha(v_H - c_H) - \kappa}{\kappa + (1 - \alpha)(c_H - v_L)}\right]^{-1} < \mu_0 < \hat{\mu}$ . If instead  $p_1^n = c_H$ , only Buyer 1's off-path constraint changes, yielding  $\hat{\mu} < \mu_0 < \mu^\dagger$ .

Finally, if  $p_2^n(t) < c_L$ , post-trade acquisition requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) - \kappa > 0$ , and offer optimality requires  $\mu_2^t < \frac{c_H - v_L}{v_H - v_L}$ . All other conditions are unchanged, yielding  $\underline{\mu} < \mu_0 < \mu^\dagger$ . Combining these outcome-equivalent profiles, Case C exists iff  $\underline{\mu} < \mu_0 < \mu^\dagger$  (non-empty when  $\kappa < \kappa^m := \frac{(2\alpha - 1)(v_H - c_H)(c_H - v_L)}{v_H - v_L}$ ) and  $\delta > \frac{\bar{\delta}}{1 - \alpha}$ .

D. *Buyer 2 mixes over acquisition after trade and acquires after no trade.* (corresponds to region  $R_4^M$ )

Buyer 2's off-path no-acquisition offer after no trade may be either  $c_H$  or some  $p < c_L$ ; the resulting profiles are outcome-equivalent. First suppose  $p_2^n(nt) = c_H$ .

Buyer 2's indifference after trade requires  $\mu_2^t(v_H - c_H) + (1 - \mu_2^t)(v_L - c_H) = \alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) - \kappa$ , so  $\mu_2^t = \bar{\mu}$  and  $\eta_D = \left(\frac{\mu_0}{1 - \mu_0}\right) \left(\frac{1 - \bar{\mu}}{\bar{\mu}}\right) - \frac{1 - \alpha}{\alpha}$ , with  $0 < \eta_D < 1$ .

After no trade, acquisition must be optimal, so  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) - \kappa > \mu_2^{nt}(v_H - c_H) + (1 - \mu_2^{nt})(v_L - c_H)$ , while  $p_2^n(nt) = c_H$  requires  $\mu_2^{nt} > \hat{\mu}$ .

Type  $L$ 's indifference at  $v_L$  gives  $(v_L - c_L) + \delta[\beta_D(1 - \alpha) + (1 - \beta_D)](c_H - c_L) = \delta(1 - \alpha)(c_H - c_L)$ , so  $\beta_D = 1 - \frac{c_L - v_L}{\delta\alpha(c_H - c_L)}$ , with  $0 < \beta_D < 1$  if and only if  $\delta > \frac{\bar{\delta}}{\alpha}$ . The remaining conditions yield  $\left[1 + (1 - \alpha)\frac{v_H - c_H}{c_H - v_L} + \alpha\frac{(1 - \alpha)(v_H - c_H) + \kappa}{\alpha(c_H - v_L) - \kappa}\right]^{-1} < \mu_0 < \bar{\mu}$ .

If instead  $p_2^n(nt) < c_L$ , no-trade acquisition requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) - \kappa > 0$ , and offer optimality requires  $\mu_2^{nt} < \frac{c_H - v_L}{v_H - v_L}$ . All other conditions are unchanged, yielding  $\mu^\dagger < \mu_0 < \left[1 + (1 - \alpha)\frac{v_H - c_H}{c_H - v_L} + \alpha\frac{(1 - \alpha)(v_H - c_H) + \kappa}{\alpha(c_H - v_L) - \kappa}\right]^{-1}$ .

Combining these outcome-equivalent profiles, Case D exists if and only if  $\mu^\dagger < \mu_0 < \bar{\mu}$  (non-empty when  $\kappa < \kappa^m$ ) and  $\delta > \frac{\bar{\delta}}{\alpha}$ .

**Welfare.** Expected total surplus is computed case by case, as in the opaque benchmark; the algebra is omitted for brevity. The ex-ante surplus levels are:

$$\begin{aligned}
ETSA &= \mu_0[(v_H - c_H)(\alpha + \delta\alpha(1 - \beta_A + \alpha\beta_A)) - \kappa(1 + \delta\alpha\beta_A)] \\
&\quad + (1 - \mu_0)[(v_L - c_L)((1 - \alpha) + \alpha\eta_A)(1 + \delta(1 - \alpha\beta_A)) - \kappa(1 + \delta\beta_A((1 - \alpha) + \alpha\eta_A))], \\
ETSB &= \mu_0[(v_H - c_H)(\alpha(1 + \delta) + \alpha(1 - \alpha)\delta\beta_B) - \kappa(1 + (1 - \alpha)\delta\beta_B)] \\
&\quad + (1 - \mu_0)[(v_L - c_L)((1 + \delta)((1 - \alpha) + \alpha\eta_B) + \alpha(1 - \alpha)(1 - \eta_B)\delta\beta_B) - \kappa(1 + \alpha(1 - \eta_B)\delta\beta_B)], \\
ETSC &= \mu_0[(\alpha(v_H - c_H) - \kappa)(1 + \delta(\alpha + \beta_C - \alpha\beta_C))] \\
&\quad + (1 - \mu_0)[(v_L - c_L)((1 + \delta) + \alpha(\eta_C - 1 - \delta) + \alpha(\alpha - 1)\delta(1 - \beta_C)(1 - \eta_C)) \\
&\quad + \kappa(-(1 + \delta) + \alpha\delta(1 - \beta_C)(1 - \eta_C))], \\
ETSD &= \mu_0[(v_H - c_H)(\alpha + \delta\alpha(2 - \alpha - (1 - \alpha)\beta_D)) - \kappa(1 + \delta(1 - \alpha + \alpha\beta_D))] \\
&\quad + (1 - \mu_0)[(v_L - c_L)(1 - \alpha + \alpha\eta_D + \delta(1 - \alpha\beta_D - \alpha^2 + \alpha^2\beta_D + \alpha^2\eta_D - \alpha^2\beta_D\eta_D)) \\
&\quad - \kappa(1 - \delta(\alpha\beta_D + \alpha\eta_D - \alpha - \beta_D - \alpha\beta_D\eta_D))].
\end{aligned}$$

Comparing these expressions with the opaque benchmark surplus yields  $ETSC < ETS_{\text{opaque}}$  and  $ETSD > ETS_{\text{opaque}}$  directly. For Case A,  $ETSA > ETS_{\text{opaque}} \iff \mu_0 < \mu_A^{\text{welf}}$ , so the welfare ranking is ambiguous in general. Moreover,  $\mu_A^{\text{welf}} < \mu^\dagger \iff \kappa < \kappa_{m,A}(\alpha, \delta)$ .<sup>10</sup> Hence, under the additional restriction  $\kappa < \kappa_{m,A}(\alpha, \delta)$ ,  $ETSA < ETS_{\text{opaque}}$  throughout

<sup>10</sup>Let  $M_A \equiv \delta(v_H - c_H) + (c_L - v_L)(1 + \delta)$  and  $N_A \equiv (v_H - c_H)[\delta(c_H - c_L) - (c_L - v_L)]$ . Then  $\kappa_{m,A} =$

Case A. For Case B,  $ETS_B > ETS_{\text{opaque}} \iff \mu_0 > \mu_B^{\text{welf}}$ , and  $\mu^\dagger > \mu_B^{\text{welf}}$ . Hence,  $ETS_B > ETS_{\text{opaque}}$  throughout the equilibrium region  $B$ . The regions  $R_1^M - R_4^M$  in Figure 1 correspond to Cases A–D, respectively. This completes the proof.  $\square$

**Lemma C.5** (Exhaustiveness). *Generically in  $(\mu_0, \delta)$ , the equilibria characterized in  $R_1^P - R_3^P$  and  $R_1^M - R_4^M$  exhaust the set of PBEs in which Buyer 1 acquires his private signal.*

*Proof.* Consider first a pure-strategy PBE. By Lemma C.2, the period-1 trade outcome perfectly reveals Buyer 1's signal. Claim 2 of the pure-strategy characterization then implies  $(d_2(t), d_2(nt)) \in \{(a, n), (n, n), (n, a)\}$ , which are precisely the configurations underlying  $R_1^P - R_3^P$ . In these three cases, the  $L$ -seller's continuation advantage  $U_t - U_{nt}$  equals  $(1 - \alpha)(c_H - c_L)$ ,  $c_H - c_L$ , and  $\alpha(c_H - c_L)$ , respectively. Since  $p_1^* = c_L - \delta(U_t - U_{nt})$  and a pure-strategy equilibrium requires  $p_1^* \geq v_L$ , the corresponding bounds are  $\delta \leq \bar{\delta}/(1 - \alpha)$ ,  $\delta \leq \bar{\delta}$ , and  $\delta \leq \bar{\delta}/\alpha$ . Hence no additional pure-strategy PBE can arise.

Now consider a mixed-strategy PBE. By Lemma C.3, the  $L$ -seller can mix only at  $p_1^* = v_L$ . Moreover, Lemma C.4 shows that, generically, Buyer 2 may randomize only over acquisition, must randomize jointly with the  $L$ -seller, and can randomize after at most one period-1 outcome. Thus every mixed-strategy PBE must belong to one of the four configurations  $R_1^M - R_4^M$ . The seller and Buyer 2 indifference conditions, together with feasibility of the mixing probabilities, yield exactly the parameter restrictions derived for those regions.

Therefore, apart from nongeneric boundary values, every PBE with information acquisition by Buyer 1 belongs to one of  $R_1^P - R_3^P$  or  $R_1^M - R_4^M$ .  $\square$

#### C.1.4 Proof of Theorem 1

The proof follows from Lemma C.5 and the pure- and mixed-strategy analyses. The first part of the theorem follows directly from the equilibrium strategies of Buyer 2 defined and proved above. It only remains to define  $\kappa^*$ . Let  $\kappa^* \equiv \kappa^{\text{welf}}(\alpha)$ , where  $\kappa^{\text{welf}}(\alpha)$  is defined in (A.3), and suppose  $0 \leq \kappa < \kappa^*$ . Under this restriction, transparency raises welfare in  $R_2^P$ ,  $R_3^P$ ,  $R_2^M$ , and  $R_4^M$ , while it lowers welfare in  $R_3^M$ . It can be verified that  $R_3^M$  shrinks as  $\alpha$  increases because its lower boundary,  $\bar{\delta}/(1 - \alpha)$ , rises with  $\alpha$ . For  $\alpha > \tilde{\alpha} = (c_H - 2c_L + v_L)/(c_H - c_L)$ , the region disappears, and  $R_1^P$  and  $R_1^M$  occupy the corresponding parameter space instead. In  $R_1^M$ , the welfare effect is generally ambiguous. As an additional refinement, I show that transparency

$$\frac{-\alpha(1 - \alpha)M_A + \sqrt{\alpha^2(1 - \alpha)^2M_A^2 + 4\alpha(1 - \alpha)^2\delta(2\alpha - 1)N_A}}{2\delta(2\alpha - 1)}. \text{ Also, } \mu_B^{\text{welf}} = \frac{\alpha(c_L - v_L)(1 + \delta) - \delta\kappa}{\mathcal{D}_B}, \text{ where}$$

$$\mathcal{D}_B = [\alpha(v_H - c_H) - \kappa](\delta - \frac{c_L - v_L}{c_H - c_L}) + \alpha(c_L - v_L)(1 + \delta) + (\frac{\alpha(v_H - c_H) - \kappa}{\kappa + (1 - \alpha)(c_H - v_L)})[(1 - \alpha)(c_L - v_L)(1 + \frac{c_L - v_L}{c_H - c_L}) - \kappa(\delta - \frac{c_L - v_L}{c_H - c_L})].$$

lowers welfare throughout  $R_1^M$  under the stronger region-specific condition  $\kappa < \kappa_{m,A}(\alpha, \delta)$ . As discussed earlier, in  $R_1^P$ , the welfare ranking depends on  $\alpha$ . For  $\alpha < \hat{\alpha}$ ,<sup>11</sup> transparency lowers welfare throughout the equilibrium region, whereas for  $\alpha \geq \hat{\alpha}$  it raises welfare for  $\mu_0 < \mu^*$  and lowers welfare for  $\mu_0 > \mu^*$ . Combining these region-specific comparisons yields the welfare characterization stated in the theorem.  $\square$

### C.1.5 Proof of Lemma 5.1

Fix any interior signal precision  $\alpha_0 \in (1/2, 1)$  satisfying the equilibrium conditions, and choose  $\delta$  such that  $\bar{\delta}/\alpha_0 < \delta < \bar{\delta}/(1 - \alpha_0)$ . Since  $\alpha_0 < 1$ , this implies  $\delta > \bar{\delta}$ . The sign of  $\partial\mu^\dagger(\alpha)/\partial\alpha$  is the sign of  $N(\alpha, \kappa) \equiv (c_H - v_L)(v_H - c_H)(2\alpha^2 - 2\alpha + 1) - (2\alpha - 1)(v_H - v_L)\kappa + 2\kappa^2$ . Because  $N(\alpha_0, 0) > 0$ , continuity implies that there exists  $\bar{\kappa}_N(\alpha_0) > 0$  such that  $\mu^\dagger(\alpha)$  is locally increasing at  $\alpha_0$  whenever  $0 < \kappa < \bar{\kappa}_N(\alpha_0)$ . Moreover,  $\kappa_{m,A}(\alpha_0, \delta) > 0$  under the stated patience condition. Hence  $\kappa$  can be chosen sufficiently small so that  $\mu^\dagger$  is locally increasing at  $\alpha_0$ , all relevant equilibrium conditions hold, and  $\kappa < \kappa_{m,A}(\alpha_0, \delta)$ .

Choose  $\mu_0$  arbitrarily close to  $\mu^\dagger(\alpha_0)$  from above, with  $\mu^\dagger(\alpha_0) < \mu_0 < \bar{\mu}(\alpha_0)$ . Then equilibrium  $D$  is active at  $\alpha_0$ , and  $\Delta_D(\alpha_0) \equiv ETS_D - ETS_{\text{opaque}} = (1 - \mu_0)(1 - \alpha_0)(c_L - v_L) > 0$ . Holding  $\mu_0$ ,  $\delta$ , and  $\kappa$  fixed, increase  $\alpha$ . Since  $\mu^\dagger$  is locally increasing at  $\alpha_0$  and  $\mu_0$  can be chosen arbitrarily close to  $\mu^\dagger(\alpha_0)$ , there exists  $\alpha^* > \alpha_0$ , arbitrarily close to  $\alpha_0$ , such that  $\mu^\dagger(\alpha^*) = \mu_0$ . By continuity, the relevant equilibrium inequalities and the welfare condition  $\kappa < \kappa_{m,A}(\alpha, \delta)$  continue to hold in a neighborhood of  $\alpha^*$ . Hence, for  $\alpha > \alpha^*$  sufficiently close to  $\alpha^*$ , we have  $\mu^\dagger(\alpha) < \mu_0 < \mu^\dagger(\alpha)$ , so equilibrium  $A$  is active, and the negative-welfare result for equilibrium  $A$  applies.

At the common boundary,  $\Delta_A = -(1 - \alpha)[\delta(c_H - c_L) - (c_L - v_L)][\alpha(1 - \alpha)(c_H - v_L)(v_H - c_H) + \kappa^2]/(c_H - v_L)[\alpha(1 - \alpha)(v_H - v_L) + \kappa(2\alpha - 1)]$ . Since  $\delta > \bar{\delta} = (c_L - v_L)/(c_H - c_L)$ , we have  $\Delta_A < 0$  at  $\alpha = \alpha^*$ . By continuity,  $\Delta_A < 0$  also for  $\alpha > \alpha^*$  sufficiently close to  $\alpha^*$ . Therefore, taking  $\alpha_1 = \alpha_0$  and any such  $\alpha_2 > \alpha^*$ , we have  $\alpha_1 < \alpha_2$ ,  $\Delta_D(\alpha_1) > 0$ , and  $\Delta_A(\alpha_2) < 0$ . Thus, holding  $\mu_0$ ,  $\delta$ , and  $\kappa$  fixed, an increase in signal precision can reverse the welfare effect of trade transparency from positive to negative.  $\square$

## C.2 Gains from trade with both types

This subsection first proves Theorem 2. I begin with Lemma 4.2, which is used to characterize the equilibria which subsequently prove the theorem. I subsequently prove several additional lemmas. Throughout, the condition  $c_H - v_L > \frac{1-\alpha}{\alpha^2}(c_H - c_L)$ , is imposed which ensures that the full equilibrium classification applies.

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<sup>11</sup>It can be easily verified that  $\hat{\alpha} > \tilde{\alpha}$ .

### C.2.1 Proof of Lemma 4.2

*Proof. Step 1: Opaque market.* The acquisition region in the opaque benchmark is derived in the proof of Proposition 1 to be  $\underline{\mu} < \mu_0 < \tilde{\mu}$ .

**Step 2: Transparent market.** Let  $p_1^*$  denote the low-quality seller's period-1 reservation price. The acquisition argument is identical to that in the proof of Proposition 1, with  $c_L$  replaced by  $p_1^*$ . Hence, Buyer 1 acquires iff  $\underline{\mu}^{p_1^*} := \frac{(1-\alpha)(c_H-p_1^*)+\kappa}{(1-\alpha)(c_H-p_1^*)+\alpha(v_H-c_H)} < \mu_0 < \frac{\alpha(c_H-p_1^*)-\kappa}{\alpha(c_H-p_1^*)+(1-\alpha)(v_H-c_H)} =: \bar{\mu}^{p_1^*}$ . For sufficiently low  $\kappa$  and  $p_1^* \in (c_L, v_L]$ , straightforward algebra gives  $\underline{\mu}^{p_1^*} \in [\underline{\mu}, \tilde{\mu})$  and  $\bar{\mu}^{p_1^*} \in [\bar{\mu}, \tilde{\mu})$ . The equilibrium analyses below establish that  $p_1^*$  lies in this interval. Combining the acquisition regions in the two markets yields the result.  $\square$

### C.2.2 Pure-strategy equilibria

The following lemma is needed for the proof of equilibria in pure strategies.

**Lemma C.6.** *In any pure-strategy equilibrium in which Buyer 1 acquires information, the low-quality seller's period 1 reservation price satisfies  $p_1^* \leq v_L$ , and she trades in period 1.*

*Proof.* Suppose, contrary to the first claim, that  $p_1^* > v_L > c_L$ . Since  $p_1^* = c_L + \delta(U_{nt} - U_t)$ <sup>12</sup>, where  $U_{nt}$  and  $U_t$  are the low-quality seller's continuation values after no trade and trade, respectively, this implies  $U_{nt} > U_t$ . Buyer 1 never offers weakly above  $p_1^*$  after a low signal, so on-path trade can occur only after a high signal and therefore reveals Buyer 1's signal. Buyer 2's interim belief is consequently weakly higher after trade than after no trade, an ordering preserved even if Buyer 2 acquires his own signal. Hence  $U_t \geq U_{nt}$ , a contradiction. Therefore,  $p_1^* \leq v_L$ . It remains to show that the low-quality seller trades in period 1. If  $p_1^* < v_L$  and the seller did not trade, Buyer 1 could profitably offer  $p_1^* + \varepsilon$ , where  $\varepsilon > 0$  is small enough that  $p_1^* + \varepsilon < v_L$ . The offer would be accepted and would give Buyer 1 positive surplus from trade with the low-quality seller. If instead  $p_1^* = v_L$  and the seller rejected an offer of  $v_L$ , trade would again reveal a high signal and no trade a low signal, implying  $U_t \geq U_{nt}$ . But  $p_1^* = c_L + \delta(U_{nt} - U_t) = v_L > c_L$  implies  $U_{nt} > U_t$ , a contradiction. Thus, the low-quality seller trades in period 1.  $\square$

**Equilibrium behavior of the players** As for the  $v_L < c_L$  case, I construct and characterize the set of pure-strategy equilibrium profiles. Define  $\tilde{\delta} := (v_L - c_L)/(c_H - c_L)$  and the

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<sup>12</sup>by the definition of reservation price

following threshold beliefs for the buyers  $\underline{\mu}^{p_1^*} < \underline{\mu} < \mu' < \mu'' < \bar{\mu}^{p_1^*}$ :

$$\begin{aligned}
\underline{\mu}^{p_1^*} &:= \frac{\kappa + (1 - \alpha)(c_H - p_1^*)}{\alpha(v_H - c_H) + (1 - \alpha)(c_H - p_1^*)} & \underline{\mu} &= \frac{\kappa + (1 - \alpha)(c_H - c_L)}{\alpha(v_H - c_H) + (1 - \alpha)(c_H - c_L)}, \\
\mu'' &:= \left(1 + \alpha \frac{\alpha(v_H - c_H) - \kappa}{(1 - \alpha)(c_H - c_L) + \kappa}\right)^{-1}, & \bar{\mu}^{p_1^*} &:= \frac{\alpha(c_H - p_1^*) - \kappa}{\alpha(c_H - p_1^*) + (1 - \alpha)(v_H - c_H)}, \\
\tilde{\mu} &:= \frac{\alpha(c_H - c_L) - \kappa}{\alpha(c_H - c_L) + (1 - \alpha)(v_H - c_H)}, \\
\mu' &= \left(1 + \alpha \frac{\alpha(v_H - c_H) - \kappa}{(1 - \alpha)(c_H - c_L) + \kappa} + (1 - \alpha) \frac{(1 - \alpha)(v_H - c_H) + \kappa}{\alpha(c_H - c_L) - \kappa}\right)^{-1}.
\end{aligned} \tag{A.4}$$

The common acquisition region for  $\mu_0$ , as defined in Lemma 4.2, is  $\underline{\mu} < \mu_0 < \bar{\mu}^{p_1^*}$ . Throughout the following proof, **I restrict attention to this region**, leaving the remaining cases to be addressed separately. I first establish the following result concerning Buyer 1's equilibrium behavior.

**Claim 1.** Given posterior belief  $\mu_1$ , Buyer 1 offers  $c_H$  if and only if  $\mu_1 \geq \frac{c_H - p_1^*}{v_H - p_1^*}$ , and otherwise offers  $p_1^*$ . He acquires his private signal if and only if  $\underline{\mu}^{p_1^*} < \mu_0 < \bar{\mu}^{p_1^*}$ , in which case he offers  $c_H$  after a high signal and  $p_1^*$  after a low signal.

*Proof.* By Lemma C.6,  $p_1^* \leq v_L$ . The offer rule follows by comparing the payoffs from  $p_1^*$  and  $c_H$ . The acquisition condition follows from Proposition 1 after replacing  $p_L$  by  $p_1^*$  and  $\hat{\mu}$  by  $\frac{c_H - p_1^*}{v_H - p_1^*}$ .  $\square$

Then I characterize behavior common to all pure-strategy equilibria.

**Claim 2.** In any pure-strategy equilibrium:

1. The high-quality seller accepts in either period if and only if  $p \geq c_H$ .
2. The low-quality seller accepts in period 1 if and only if  $p_1 \geq p_1^*$  and in period 2 if and only if  $p_2 \geq c_L$ .
3. Buyer 1 acquires his signal and offers  $p_1^h = c_H$  and  $p_1^l = p_1^*$ .
4. Buyer 2 offers  $c_H$  after a high signal and  $c_L$  after a low signal if he acquires, and offers  $c_H$  after no trade if he does not.

*Proof.* The high-quality seller's behavior follows from Lemma A.1. By Lemma C.6, the low-quality seller trades in period 1, so she accepts  $p_1^*$  on path; her period 2 behavior is the same as under opacity. Claim 1 and the common acquisition-region restriction together imply  $\underline{\mu} < \mu_0 < \bar{\mu}^{p_1^*}$ . Claim 1 also determines Buyer 1's off-path no-acquisition offer:  $p_1^n = c_H$  if and only if  $\mu_0 > \frac{c_H - p_1^*}{v_H - p_1^*}$ , and  $p_1^n = p_1^*$  otherwise. By Lemma C.1, whenever Buyer 2 acquires information, he offers  $c_H$  after a high signal and  $c_L$  after a low signal. Because the low-quality seller trades with probability one on path, no trade reveals high quality; Buyer 2 therefore

does not acquire and offers  $c_H$ .  $\square$

Thus, the only remaining distinction across the pure-strategy equilibria is Buyer 2's no-acquisition offer after trade,  $p_2^n(t) \in \{c_H, c_L\}$ . Building on Claim 2, I now derive the equilibrium conditions for the pure-strategy profiles corresponding to regions  $r_1^P$  and  $r_2^P$ , respectively. In any such equilibrium, Buyer 2's posteriors are  $\mu_2^t = \frac{\alpha\mu_0}{1-(1-\alpha)\mu_0}$  after trade and  $\mu_2^{nt} = 1$  after no trade. Alternative or unsustainable off-path actions are noted within each case.

A. *Buyer 2 does not acquire after either period-1 outcome. (corresponds to region  $r_1^P$ )*

After trade, Buyer 2 may offer either  $p_2^n(t) = c_H$  or  $p_2^n(t) = c_L$ , while Buyer 1's off-path no-acquisition offer may be  $p_1^n = c_H$  or  $p_1^n = p_1^*$ . First consider  $p_2^n(t) = c_L$  and  $p_1^n = p_1^*$ . The low-quality seller's reservation price is  $p_1^* = c_L + \delta(c_H - c_L)$ .<sup>13</sup> By Lemma C.6,  $p_1^* \leq v_L$ , or equivalently  $\delta \leq \frac{v_L - c_L}{c_H - c_L}$ .

After trade, Buyer 2 does not acquire and offers  $c_L$ . This requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa < (1 - \mu_2^t)(v_L - c_L)$  and  $\mu_2^t < \frac{c_H - c_L}{v_H - c_L}$ . Together with Buyer 1's constraints, these conditions imply  $\mu < \mu_0 < \mu''$ . For  $\kappa < \kappa_{p,A}$ , where  $\kappa_{p,A} \equiv \frac{(v_H - c_H)(c_H - c_L)[\alpha^2(1 - \delta) - (1 - \alpha)]}{(v_H - c_H) + \alpha(1 - \delta)(c_H - c_L)}$ , Buyer 2's acquisition constraint is the binding upper bound on  $\mu_0$ .<sup>14</sup>

The outcome-equivalent profile with  $p_1^n = c_H$  cannot be sustained because Buyer 1's resulting off-path offer constraint,  $\mu_0 > \frac{c_H - p_1^*}{v_H - p_1^*}$ , is incompatible with Buyer 2's acquisition incentive constraint. Likewise, the profile with  $p_2^n(t) = c_H$  is ruled out because Buyer 2's no-acquisition constraint is incompatible with Buyer 1's acquisition constraint for small  $\kappa$ . Hence, Case A exists iff  $\mu < \mu_0 < \mu''$ ,  $\delta \leq \tilde{\delta}$ , and  $\kappa < \kappa_{p,A}$ .

B. *Buyer 2 acquires after trade and does not acquire after no trade. (corresponds to region  $r_2^P$ )*

Define  $\kappa_{p,B1} \equiv \frac{(v_H - c_H)(c_H - c_L)[\alpha^2(1 - \delta\alpha) - (1 - \alpha)]}{(c_H - c_L) + \alpha(v_H - c_H)}$  and  $\kappa_{p,B2} \equiv \frac{(v_H - c_H)(c_H - c_L)[\alpha^2(1 - \delta\alpha) - (1 - \alpha)]}{(v_H - c_H) + \alpha(1 - \delta\alpha)(c_H - c_L)}$ .

Buyer 2's possible off-path no-acquisition offers after trade are  $p_2^n(t) \in \{c_L, c_H\}$ , while Buyer 1's off-path offer satisfies  $p_1^n \in \{c_H, p_1^*\}$ . First consider  $p_2^n(t) = c_H$  and  $p_1^n = c_H$ . The low-quality seller's reservation price is  $p_1^* = c_L + \delta\alpha(c_H - c_L)$ .<sup>15</sup> By Lemma C.6,  $p_1^* \leq v_L$ , or equivalently  $\delta \leq \frac{v_L - c_L}{\alpha(c_H - c_L)}$ .

<sup>13</sup>Her indifference condition is  $p_1^* - c_L + \delta(0) = 0 + \delta(c_H - c_L)$ .

<sup>14</sup>The threshold  $\kappa_{p,A}$  is strictly positive under the maintained conditions  $\delta < \frac{v_L - c_L}{c_H - c_L}$  and  $c_H - v_L > \frac{1 - \alpha}{\alpha^2}(c_H - c_L)$ ; and also decreases in  $\delta$ .

<sup>15</sup>Her indifference condition is  $p_1^* - c_L + \delta(1 - \alpha)(c_H - c_L) = \delta(c_H - c_L)$ .

After observing trade in period 1, acquisition by Buyer 2 requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa > \mu_2^t(v_H - c_H) + (1 - \mu_2^t)(v_L - c_H)$ , while sequential rationality of the off-path offer requires  $\mu_2^t > \hat{\mu} = \frac{c_H - c_L}{v_H - c_L}$ . Together with Buyer 1's constraints, these conditions sustain the profile if and only if  $(1 + \alpha \frac{v_H - c_H}{c_H - c_L})^{-1} < \mu_0 < \bar{\mu}^{p_1^*}$  and  $\kappa < \kappa_{p,B1}$ .

For the outcome-equivalent profile with  $p_1^n = p_1^*$ , Buyer 1's off-path sequential-rationality constraint becomes  $\mu_0 < \frac{c_H - p_1^*}{v_H - p_1^*}$ . The profile is not sustainable because this constraint contradicts the constraint corresponding to Buyer 2's belief upon observing trade.

Two further outcome-equivalent profiles have  $p_2^n(t) = c_L$  and either  $p_1^n = c_H$  or  $p_1^n = p_1^*$ . Only Buyer 2's constraints change: acquisition requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa > (1 - \mu_2^t)(v_L - c_L)$ , and sequential rationality of the off-path offer requires  $\mu_2^t < \hat{\mu}$ . These profiles are sustained iff  $\frac{c_H - p_1^*}{v_H - p_1^*} < \mu_0 < (1 + \alpha \frac{v_H - c_H}{c_H - c_L})^{-1}$ ;  $\kappa < \min\{\kappa_{p,B1}, \kappa_{p,B2}\}$  and  $(1 + \alpha \frac{\alpha(v_H - c_H) - \kappa}{(1 - \alpha)(c_H - c_L) + \kappa})^{-1} < \mu_0 < \frac{c_H - p_1^*}{v_H - p_1^*}$ ;  $\kappa < \kappa_{p,B2}$ , respectively. Combining all outcome-equivalent profiles, Case B exists if and only if  $\mu'' < \mu_0 < \bar{\mu}^{p_1^*}$ ;  $\delta \leq \frac{\tilde{\delta}}{\alpha}$  and  $\kappa < \kappa_{p,B} \equiv \min\{\kappa_{p,B1}, \kappa_{p,B2}\}^{16}$ .

**Welfare.** I now compare social surplus in Cases A and B with that under opacity. Ex-ante expected total surplus for the respective profiles is (denoted by subscript):

$$\begin{aligned} ETS_A &= \mu_0 \left[ (\alpha + \delta(1 - \alpha))(v_H - c_H) - \kappa \right] + (1 - \mu_0) \left[ v_L - c_L - \kappa + \delta(v_L - c_L) \right], \\ ETS_B &= \mu_0 \left[ (v_H - c_H)(\alpha + \delta(1 - \alpha + \alpha^2)) - \kappa(1 + \alpha\delta) \right] + (1 - \mu_0)(v_L - c_L - \kappa)(1 + \delta). \end{aligned}$$

Straightforward algebra yields<sup>17</sup>  $ETS_A > ETS_{\text{opaque}}^a$  if and only if  $\mu_0 < \frac{\kappa}{(2\alpha - 1)(v_H - c_H)}$ . For this inequality to hold anywhere in Case A, the cutoff must exceed the region's lower bound, which requires  $\kappa > \kappa^m \equiv \frac{(2\alpha - 1)(v_H - c_H)(c_H - v_L)}{v_H - v_L}$ . This is incompatible with the feasibility of Case A since  $\kappa^m > \kappa_{p,A}$ . Hence, transparency lowers surplus throughout Case A ( $r_1^P$ ). Moreover,  $ETS_B > ETS_{\text{opaque}}$ , so transparency raises surplus throughout Case B ( $r_2^P$ ).  $\square$

### C.2.3 Mixed-strategy equilibria

The following lemma is needed for mixed-strategy equilibrium characterization.

**Lemma C.7.** *In any mixed-strategy equilibrium, the low-quality seller's reservation price is  $p_1^* = v_L$ .*

<sup>16</sup>  $\kappa < \kappa_{p,B1}$  ensures that Buyer 2's belief constraint is the binding upper bound on  $\mu_0$ , while  $\kappa < \kappa_{p,B2}$  ensures that Buyer 1's off-path belief constraint is the binding lower bound.

<sup>17</sup> I use  $ETS_{\text{opaque}}^a$  as the benchmark because the common acquisition region is being studied.

*Proof.* Because Buyer 1 values the low-quality seller at  $v_L$ , he never offers a price  $p \in (v_L, c_H)$  to induce trade only with that type. Also, by Lemma A.2, the low-quality seller strictly accepts an offer of  $c_H$ . Because she strictly accepts  $c_H$ , she cannot be indifferent or mix at that price. Therefore, her reservation price cannot exceed  $v_L$ , so  $p_1^* \leq v_L$ . If  $p_1^* < v_L$ , Buyer 1 could offer some  $p' \in (p_1^*, v_L)$ , which the low-quality seller would strictly accept and the high-quality seller would reject, yielding Buyer 1 strictly positive surplus instead of the zero surplus he earns by mixing. Hence,  $p_1^* \geq v_L$ , so  $p_1^* = v_L$ .  $\square$

**Equilibrium behavior of players.** Now I construct and characterize the set of equilibrium mixed-strategy profiles. Consider the thresholds defined in (A.4). Moreover, since  $p_1^* = v_L$ ,  $\bar{\mu}^{p_1^*} = \frac{\alpha(c_H - v_L) - \kappa}{\alpha(c_H - v_L) + (1 - \alpha)(v_H - c_H)}$ . I first prove a lemma that rules out alternative randomization patterns.

**Lemma C.8.** *Generically, every mixed-strategy PBE has the randomization structure as described in Section C.2. No other pattern of equilibrium randomization can be sustained.*

*Proof. Step 1.* Generically, Buyer 1 does not mix over information acquisition or offers.

*Proof of Step 1.* Suppose first that the low-quality seller plays a pure strategy. By Claim 1 of Lemma C.6,  $p_1^* \leq v_L$ , and Buyer 1 induces acceptance on path by offering  $p_1^*$ . His acquisition incentives therefore coincide with those in the pure-strategy PBE, so mixing over acquisition is possible only at  $\mu_0 = \underline{\mu}^{p_1^*}$  or  $\mu_0 = \bar{\mu}^{p_1^*}$ . The lower cutoff is anyways outside the common-acquisition region being considered. Mixing over  $p_1^*$  and  $c_H$  requires the pre-offer belief to equal  $\frac{c_H - p_1^*}{v_H - p_1^*}$ . After acquisition, this can occur only when  $\mu_0 < \underline{\mu}$  or  $\mu_0 > \bar{\mu}^{p_1^*}$ , where acquisition is suboptimal; without acquisition, it requires  $\mu_0 = \frac{c_H - p_1^*}{v_H - p_1^*}$ , where acquisition is optimal. Either case contradicts equilibrium. If the low-quality seller mixes, Lemma C.7 implies  $p_1^* = v_L$ , and the same argument applies.  $\square$

**Step 2.** If Buyer 2 mixes in equilibrium, he does so only over information acquisition, not over offers.

*Proof of Step 2.* The argument parallels Step 2 of Lemma C.4. Buyer 2's offer problem is again the same as under opacity, with his interim belief replacing the prior. Mixing between  $c_H$  and  $c_L$  requires the offer-stage belief to equal  $\hat{\mu}$ . By Lemma C.1, if this occurs after acquisition, acquisition is suboptimal; if it occurs without acquisition, acquisition is optimal. Thus offer-stage indifference is incompatible with Buyer 2's acquisition incentives, so any equilibrium mixing by Buyer 2 must be over acquisition.  $\square$

**Step 3.** Generically, Buyer 2 and the low-quality seller mix simultaneously.

*Proof of Step 3.* Suppose the  $L$ -seller mixes while Buyer 2 uses a pure acquisition strategy. Her indifference condition is  $v_L - c_L + \delta U_t = \delta U_{nt}$ . By Lemma C.1, Buyer 2's signal-contingent

offers are fixed. Evaluating the indifference condition across the four possible acquisition profiles shows that it can hold only at  $\delta \in \{\tilde{\delta}, \tilde{\delta}/\alpha, \tilde{\delta}/(1-\alpha)\}$ . Hence, if Buyer 2 uses a pure acquisition strategy, the  $L$ -seller can mix only at these boundary values of  $\delta$ . If Buyer 2 mixes while the low-quality seller is pure, the period-1 outcome pins down his interim belief. After no trade,  $\mu_2^{nt} = 1$ , so acquisition is strictly suboptimal; after trade, acquisition indifference requires  $\mu_2^t \in \{\tilde{\mu}, \underline{\mu}\}$ , which occurs only at cutoff values of  $\mu_0$ . Hence, away from the identified boundary values of  $\delta$  and  $\mu_0$ , Buyer 2 and the low-quality seller must mix simultaneously.  $\square$

**Step 4.** Generically, Buyer 2 mixes over acquisition after exactly one period-1 outcome.

*Proof.* The argument parallels Step 4 of Lemma C.4. Suppose Buyer 2 mixes after both trade and no trade. By Step 3, the low-quality seller also mixes. Buyer 2's two interim beliefs must each equal one of the acquisition cutoffs,  $\tilde{\mu}$  or  $\underline{\mu}$ . As in the corresponding argument for  $v_L < c_L$ , assigning the same cutoff to both beliefs yields a boundary prior, while  $\mu_2^t = \underline{\mu}$  and  $\mu_2^{nt} = \tilde{\mu}$  implies  $\mu_0 = \mu'$ . In the remaining case,  $\mu_2^t = \tilde{\mu}$  and  $\mu_2^{nt} = \underline{\mu}$ , the seller's indifference condition reduces to  $\delta(c_H - c_L)[\alpha\beta + (1-\alpha)\beta' - 1] = v_L - c_L$ . This is impossible because  $\beta, \beta' \in (0, 1)$  imply  $\alpha\beta + (1-\alpha)\beta' - 1 < 0$ , whereas  $v_L - c_L > 0$ . Hence, away from boundary priors, Buyer 2 cannot mix after both period-1 outcomes.  $\square$

Combining Steps 1–4, in any mixed-strategy PBE Buyer 2 randomizes only over acquisition, the low-quality seller randomizes simultaneously, and Buyer 2 randomizes after at most one period-1 outcome. Hence, only the four mixed-strategy configurations discussed in Section C.2 can arise.  $\square$

Lemma C.8 proves the structure of Buyer 2's acquisition behavior. The following claim lists the candidate offers by Buyer 2 on not acquiring his signal which might be sustained in a mixed-strategy equilibrium.

**Claim 1.** For Cases (A)–(D), corresponding to regions  $r_1^M - r_4^M$  in the mixed-strategy discussion in Section C.2, the candidate non-acquisition offers  $(p_2^n(t), p_2^n(nt))$  that may be consistent with equilibrium are: (A)  $(c_L, c_H)$  or  $(c_L, c_L)$ ; (B)  $(c_H, c_H)$  or  $(c_L, c_H)$ ; (C)  $p_2^n(t) = c_L$ ; (D)  $p_2^n(nt) = c_H$ .

*Proof.* Buyer 2's only relevant no-acquisition offers are  $c_H$ , which is accepted by both seller types, and  $c_L$ , which is accepted only by the low-quality seller. Substituting each offer pair and the acquisition behavior in Cases A–D into the low-quality seller's indifference condition eliminates all combinations not listed above.<sup>18</sup>  $\square$

Now I describe the equilibrium behavior of the players common across mixed equilibria.

**Claim 2.** The following behavior is common to all mixed-strategy PBEs: [-2.5em]

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<sup>18</sup>The continuation values after trade and no trade also depend on Buyer 2's acquisition probability. Conditional on acquisition, his offers are pinned down by Lemma C.1.

1. The high-quality seller accepts in either period if and only if  $p \geq c_H$ .
2. The low-quality seller mixes between accepting and rejecting  $v_L$  in period 1 and accepts in period 2 if and only if  $p_2 \geq c_L$ .
3. Buyer 1 acquires his signal and offers  $p_1^h = c_H$  and  $p_1^l = v_L$ .
4. Upon acquisition, Buyer 2 offers  $c_H$  after a high signal and  $c_L$  after a low signal. His no-acquisition behavior is described in Section C.2.

*Proof.* The high-quality seller's behavior follows from Lemma A.1, and the low-quality seller's behavior from Lemma C.7. As in Claim 1 for the pure-strategy equilibrium, with  $p_1^*$  replaced by  $v_L$ , Buyer 1 acquires if and only if  $\underline{\mu}^{p_1^*} < \mu_0 < \bar{\mu}^{p_1^*}$ . His equilibrium offers are  $p_1^h = c_H$  and  $p_1^l = v_L$ : after a high signal, he must induce acceptance by the high-quality seller, since otherwise acquisition would be strictly less valuable, while Lemma C.7 implies that  $v_L$  is the only sustainable period-1 offer below  $c_H$ . Following a deviation to no acquisition, sequential rationality requires  $p_1^n = c_H$  if and only if  $\mu_0 > \frac{c_H - v_L}{v_H - v_L}$ , and  $p_1^n = v_L$  otherwise. Buyer 2's post-acquisition offers follow from Lemma C.1.  $\square$

I now derive the equilibrium conditions case by case for Cases A–D. In each case, Buyer 2 acquires with probability  $\beta$ , all other strategies follow Claim 2, and the relevant acquisition and no-acquisition offer constraints are imposed. If the low-quality seller accepts with probability  $\eta$ , Buyer 2's posteriors are  $\mu_2^t = \frac{\alpha\mu_0}{\alpha\mu_0 + (1-\alpha+\alpha\eta)(1-\mu_0)}$  and  $\mu_2^{nt} = \frac{(1-\alpha)\mu_0}{(1-\alpha)\mu_0 + \alpha(1-\eta)(1-\mu_0)}$ .

A. *Buyer 2 mixes over acquisition after no trade and does not acquire after trade.* (corresponds to region  $r_1^M$ )

Define  $\kappa_{m,A} \equiv \frac{C - \sqrt{C^2 - 4(v_H - v_L)F}}{2(v_H - v_L)}$ , where  $C = (2\alpha - 1)[(v_H - c_H)(c_H - c_L) + (v_H - c_H)(c_H - v_L) + (c_H - c_L)(c_H - v_L)]$  and  $F = (v_H - c_H)(c_H - c_L)[(c_H - v_L)(3\alpha^2 - 3\alpha + 1) - \alpha(1 - \alpha)(c_H - c_L)]$ , and assume  $\kappa < \kappa_{m,A}$  throughout this profile. By Claim 1, Buyer 2 has two feasible on-path no-acquisition offer profiles.

**Case 1:**  $(p_2^n(t), p_2^n(nt)) = (c_L, c_H)$ , and Buyer 1's off-path offer is  $p_1^n = v_L$ . Buyer 2's indifference after no trade requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) + \alpha(1 - \mu_2^{nt})(v_L - c_L) - \kappa = \mu_2^{nt}(v_H - c_H) + (1 - \mu_2^{nt})(v_L - c_H)$ , which yields  $\eta_A = 1 - \frac{1-\alpha}{\alpha} \frac{\mu_0}{1-\mu_0} \frac{1-\tilde{\mu}}{\tilde{\mu}}$ , with  $0 < \eta_A < 1$ . After trade, not acquiring and offering  $c_L$  requires  $\mu_2^t < \hat{\mu}$  and  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa < (1 - \mu_2^t)(v_L - c_L)$ .

The low-quality seller's indifference condition is  $(v_L - c_L) = \delta[\beta_A(1 - \alpha) + (1 - \beta_A)](c_H - c_L)$ , so  $\beta_A = \frac{1}{\alpha} \left(1 - \frac{v_L - c_L}{\delta(c_H - c_L)}\right)$ . Feasibility requires  $0 < \beta_A < 1$ , or equivalently  $\tilde{\delta} < \delta < \frac{\tilde{\delta}}{1 - \alpha}$ . Together with Buyer 1's constraints, these conditions imply  $\mu < \mu_0 < \mu'$  and  $\tilde{\delta} < \delta < \frac{\tilde{\delta}}{1 - \alpha}$ , with Buyer 1's acquisition constraint binding at the lower bound.

For the outcome-equivalent profile with  $p_1^n = c_H$ , only Buyer 1's off-path constraint changes, becoming  $\mu_0 > \frac{c_H - v_L}{v_H - v_L}$ . Under  $\kappa < \kappa_{m,A}$ , this constraint is incompatible with Buyer 2's no-acquisition constraint after trade. Hence, Case 1 exists iff  $\mu < \mu_0 < \mu'$ ,  $\tilde{\delta} < \delta < \frac{\tilde{\delta}}{1-\alpha}$ , and  $\kappa < \kappa_{m,A}$ .

**Case 2:**  $(p_2^n(t), p_2^n(nt)) = (c_L, c_L)$ .

As in Case 1, first consider  $p_1^n = c_H$ . Buyer 1's acquisition constraint and Buyer 2's acquisition constraint upon observing trade are mutually incompatible. The outcome equivalent profile with  $p_1^n = v_L$ , can't be sustained in equilibrium in the common acquisition region  $\mu < \mu_0 < \bar{\mu}^{p^1}$ . It can be sustained in equilibrium outside this region, which I will discuss later.

- B. *Buyer 2 mixes over acquisition after trade and does not acquire after no trade.* (corresponds to region  $r_2^M$ )

Define  $\kappa_{m,B} \equiv \frac{(v_H - c_H)[\alpha^2(c_H - v_L) - (1-\alpha)(c_H - c_L)]}{(v_H - c_H) + \alpha(c_H - v_L)}$  and suppose throughout this profile that  $\kappa < \kappa_{m,B}$ .<sup>19</sup> By Claim 1, Buyer 2 has two feasible on-path no-acquisition offer profiles.

**Case 1:**  $(p_2^n(t), p_2^n(nt)) = (c_L, c_H)$ , and Buyer 1's off-path offer is  $p_1^n = v_L$ . Buyer 2's indifference after trade requires  $\alpha\mu_2^t(v_H - c_H) + (1-\alpha)(1-\mu_2^t)(v_L - c_H) + \alpha(1-\mu_2^t)(v_L - c_L) - \kappa = (1-\mu_2^t)(v_L - c_L)$ , which yields  $\eta_B = \frac{\mu_0}{1-\mu_0} \frac{1-\mu}{\mu} - \frac{1-\alpha}{\alpha}$ , with  $0 < \eta_B < 1$ . After no trade, not acquiring and offering  $c_H$  requires  $\mu_2^{nt} > \hat{\mu}$  and  $\alpha\mu_2^{nt}(v_H - c_H) + (1-\alpha)(1-\mu_2^{nt})(v_L - c_H) + \alpha(1-\mu_2^{nt})(v_L - c_L) - \kappa < \mu_2^{nt}(v_H - c_H) + (1-\mu_2^{nt})(v_L - c_H)$ .

The low-quality seller's indifference condition is  $v_L - c_L + \delta\beta_B(1-\alpha)(c_H - c_L) = \delta(c_H - c_L)$ , so  $\beta_B = \frac{1}{1-\alpha} \left(1 - \frac{v_L - c_L}{\delta(c_H - c_L)}\right)$ . Feasibility requires  $0 < \beta_B < 1$ , or equivalently  $\tilde{\delta} < \delta < \tilde{\delta}/\alpha$ . Together with Buyer 1's constraints, these conditions imply  $\mu' < \mu_0 < \mu''$  and  $\tilde{\delta} < \delta < \tilde{\delta}/\alpha$ .

For the outcome-equivalent profile with  $p_1^n = c_H$ , only Buyer 1's off-path constraint changes. Under  $\kappa < \kappa_{m,B}$ , the upper bound implied by  $0 < \eta_B < 1$  lies below Buyer 1's required lower bound, so this profile cannot be sustained. Hence, Case 1 exists iff  $\mu' < \mu_0 < \mu''$  and  $\tilde{\delta} < \delta < \tilde{\delta}/\alpha$ .

**Case 2:**  $(p_2^n(t), p_2^n(nt)) = (c_H, c_H)$ . This case cannot be sustained because Buyer 2's no-trade acquisition constraint is incompatible with Buyer 1's acquisition constraint.

- C. *Buyer 2 mixes over acquisition after trade and acquires after no trade.* ( $\equiv r_3^M$ ) (corresponds to region  $r_3^M$ )

<sup>19</sup>The restriction  $\kappa < \kappa_{m,B}$  ensures that the upper bound on  $\mu_0$  implied by  $0 < \eta_B < 1$  lies below Buyer 1's off-path cutoff  $\frac{c_H - v_L}{v_H - v_L}$ .

Define  $\kappa_{m,C} = \frac{(v_H - c_H)(c_H - c_L)[(c_H - v_L)(2\alpha^2 - 2\alpha + 1) - (1 - \alpha)(c_H - c_L)]}{(v_H - c_H)(c_H - c_L) - (1 - \alpha)(v_H - c_H)(c_H - v_L) + \alpha(c_H - c_L)(c_H - v_L)}$ , and assume throughout this profile that  $\kappa < \kappa_{m,C}$ . By Claim 1,  $p_2^n(t) = c_L$ . The possible off-path offers are  $p_1^n \in \{c_H, v_L\}$  for Buyer 1 and  $p_2^n(nt) \in \{c_H, c_L\}$  for Buyer 2.

**Case 1:**  $p_1^n = v_L$  and  $p_2^n(nt) = c_H$ .

Buyer 2's indifference after trade requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa = (1 - \mu_2^t)(v_L - c_L)$ , which yields  $\eta_C = \frac{\mu_0}{1 - \mu_0} \frac{1 - \mu}{\mu} - \frac{1 - \alpha}{\alpha}$ , with  $0 < \eta_C < 1$ . After no trade, acquisition and the off-path offer  $c_H$  require  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) + \alpha(1 - \mu_2^{nt})(v_L - c_L) - \kappa > \mu_2^{nt}(v_H - c_H) + (1 - \mu_2^{nt})(v_L - c_H)$  and  $\mu_2^{nt} > \hat{\mu}$ .

The low-quality seller's indifference condition is  $(v_L - c_L) + \delta\beta_C(1 - \alpha)(c_H - c_L) = \delta(1 - \alpha)(c_H - c_L)$ , so  $\beta_C = 1 - \frac{v_L - c_L}{\delta(1 - \alpha)(c_H - c_L)}$ . Feasibility requires  $0 < \beta_C < 1$ , or  $\delta > \tilde{\delta}/(1 - \alpha)$ .

With Buyer 1's constraints, these conditions imply  $\left(1 + \alpha \frac{\alpha(v_H - c_H) - \kappa}{(1 - \alpha)(c_H - c_L) + \kappa} + (1 - \alpha) \frac{v_H - c_H}{c_H - c_L}\right)^{-1} < \mu_0 < \mu'$ , with Buyer 1's off-path constraint binding.

For the outcome-equivalent profile with  $p_1^n = c_H$ , Buyer 1's off-path constraint becomes  $\mu_0 > \frac{c_H - v_L}{v_H - v_L}$ , which is incompatible with Buyer 2's acquisition constraint after no trade. Hence this profile cannot be sustained.

**Case 2:**  $p_1^n = v_L$  and  $p_2^n(nt) = c_L$ .

Only Buyer 2's no-trade constraints change: offer optimality requires  $\mu_2^{nt} < \hat{\mu}$ , while acquisition requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) + \alpha(1 - \mu_2^{nt})(v_L - c_L) - \kappa > (1 - \mu_2^{nt})(v_L - c_L)$ . These conditions imply  $\mu < \mu_0 < \left(1 + \alpha \frac{\alpha(v_H - c_H) - \kappa}{(1 - \alpha)(c_H - c_L) + \kappa} + (1 - \alpha) \frac{v_H - c_H}{c_H - c_L}\right)^{-1}$ .

The corresponding profile with  $p_1^n = c_H$  cannot be sustained because Buyer 1's off-path constraint is incompatible with Buyer 2's belief constraint after trade.

Hence, combining the two cases, Case C exists if and only if  $\mu < \mu_0 < \mu'$ ,  $\delta > \tilde{\delta}/(1 - \alpha)$ , and  $\kappa < \kappa_{m,C}$ .

D. *Buyer 2 mixes over acquisition after no trade and acquires after trade.* (corresponds to region  $r_4^M$ )

Define  $m_D := \left(1 + \alpha \frac{v_H - c_H}{c_H - c_L} + (1 - \alpha) \frac{(1 - \alpha)(v_H - c_H) + \kappa}{\alpha(c_H - c_L) - \kappa}\right)^{-1}$  and  $x \equiv \frac{c_H - v_L}{v_H - v_L}$ . Let  $\kappa_{m,D} \equiv \min\{\kappa_{m,A}, \kappa_{m,D1}\}$ , and suppose throughout that  $\kappa < \kappa_{m,D}$ .<sup>20</sup> By Claim 1,  $p_2^n(nt) = c_H$ . Buyer 2's possible off-path no-acquisition offers after trade are  $p_2^n(t) \in \{c_H, c_L\}$ , while Buyer 1's possible off-path offers are  $p_1^n \in \{c_H, v_L\}$ .

<sup>20</sup>  $\kappa_{m,A}$  is defined in Case A.  $\kappa_{m,D1} = \frac{B - \sqrt{B^2 - 4(v_H - c_L)H}}{2(v_H - c_L)}$ , where  $B = (2\alpha - 1)(v_H - c_H)(c_H - c_L) + \alpha(v_H - c_H)(c_H - v_L) + (c_H - c_L)^2 - (1 - \alpha)(c_H - c_L)(c_H - v_L)$  and  $H = (v_H - c_H)(c_H - c_L)[(2\alpha^2 - 2\alpha + 1)(c_H - v_L) - (1 - \alpha)(c_H - c_L)]$ .

**Case 1:**  $p_1^n = c_H$  and  $p_2^n(t) = c_L$ .

Buyer 2's indifference after no trade requires  $\alpha\mu_2^{nt}(v_H - c_H) + (1 - \alpha)(1 - \mu_2^{nt})(v_L - c_H) + \alpha(1 - \mu_2^{nt})(v_L - c_L) - \kappa = \mu_2^{nt}(v_H - c_H) + (1 - \mu_2^{nt})(v_L - c_H)$ , which yields  $\eta_D = 1 - \frac{1-\alpha}{\alpha} \frac{\mu_0}{1-\mu_0} \frac{1-\tilde{\mu}}{\tilde{\mu}}$ , with  $0 < \eta_D < 1$ . After trade, acquisition and the off-path no-acquisition offer  $c_L$  require  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa > (1 - \mu_2^t)(v_L - c_L)$  and  $\mu_2^t < \hat{\mu}$ .

The low-quality seller's indifference condition is  $(v_L - c_L) + \delta(1 - \alpha)(c_H - c_L) = \delta[\beta_D(1 - \alpha) + (1 - \beta_D)](c_H - c_L)$ , so  $\beta_D = 1 - \frac{v_L - c_L}{\delta\alpha(c_H - c_L)}$ . Feasibility requires  $0 < \beta_D < 1$ , or equivalently  $\delta > \tilde{\delta}/\alpha$ . Together with Buyer 1's constraints, this profile is sustained if and only if  $x < \mu_0 < m_D$  and  $\delta > \tilde{\delta}/\alpha$ ; Buyer 1's off-path constraint is the binding lower bound.

For the outcome-equivalent profile with  $p_1^n = v_L$ , only Buyer 1's off-path constraint changes, becoming  $\mu_0 < x$ . It is sustained if and only if  $\mu' < \mu_0 < x$  and  $\delta > \tilde{\delta}/\alpha$ .

**Case 2:**  $p_1^n = c_H$  and  $p_2^n(t) = c_H$ .

Only Buyer 2's post-trade constraints change: offer optimality requires  $\mu_2^t > \hat{\mu}$ , and acquisition requires  $\alpha\mu_2^t(v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa > \mu_2^t(v_H - c_H) + (1 - \mu_2^t)(v_L - c_H)$ . All other conditions remain unchanged. Under  $\kappa < \kappa_{m,D}$ ,  $m_D$  is the binding lower bound and Buyer 1's acquisition constraint is the binding upper bound, so this profile is sustained if and only if  $m_D < \mu_0 < \bar{\mu}^{p_1^*}$  and  $\delta > \tilde{\delta}/\alpha$ .

For the outcome-equivalent profile with  $p_1^n = v_L$ , only Buyer 1's off-path constraint changes to  $\mu_0 < x$ . Since  $m_D > x$  under this restriction, while Buyer 2's belief constraint requires  $\mu_0 > m_D$ , this profile cannot be sustained.

Combining the three sustainable profiles, Case D exists if and only if  $\mu' < \mu_0 < \bar{\mu}^{p_1^*}$ ,  $\delta > \tilde{\delta}/\alpha$ , and  $\kappa < \kappa_{m,D}$ .

**Welfare.** The expected social surplus (ETS) for Cases A–D is given by:

$$\begin{aligned} ETS_A &= \mu_0 \left( (v_H - c_H) [\alpha + \delta(1 - \alpha)(1 - \beta_A(1 - \alpha))] - \kappa [1 + \beta_A \delta(1 - \alpha)] \right) \\ &\quad + (1 - \mu_0) \left( (v_L - c_L) [1 + \delta - \alpha(1 - \eta_A)] - \kappa [1 + \alpha \delta \beta_A(1 - \eta_A)] \right), \\ ETS_B &= \mu_0 \left( \alpha(v_H - c_H) - \kappa + \delta[(1 - \alpha)(v_H - c_H) + \alpha^2 \beta_B(v_H - c_H) - \alpha \beta_B \kappa] \right) \\ &\quad + (1 - \mu_0) \left( (v_L - c_L) [1 - \alpha + \delta + \alpha \eta_B] - \kappa [1 + \delta \beta_B(1 - \alpha + \alpha \eta_B)] \right), \\ ETS_C &= \mu_0 \left( (v_H - c_H) [\alpha(1 + \delta) - \alpha^2 \delta(1 - \beta_C)] - \kappa [1 + \delta - \alpha \delta(1 - \beta_C)] \right) \end{aligned}$$

$$\begin{aligned}
& + (1 - \mu_0) \left( (v_L - c_L)[1 + \delta - \alpha(1 - \eta_C)] - \kappa[1 + \delta\beta_C + \alpha\delta(1 - \beta_C)(1 - \eta_C)] \right), \\
ETS_D = & \mu_0 \left( (v_H - c_H)[\alpha + \delta(\alpha + (1 - \alpha)^2(1 - \beta_D))] - \kappa[1 + \delta(\alpha + \beta_D - \alpha\beta_D)] \right) \\
& + (1 - \mu_0) \left( (v_L - c_L)[1 + \delta - \alpha(1 - \eta_D)] - \kappa[1 + \delta - \alpha\delta(1 - \eta_D)(1 - \beta_D)] \right).
\end{aligned}$$

The feasible regions for Cases A–D coincide with  $r_1^P - r_4^P$ , respectively. Relative to  $ETS_{\text{opaque}}^a$ , transparency lowers welfare in Cases A and C and has no effect in Case D. In Case A,  $ETS_A > ETS_{\text{opaque}}^a$  would require  $\mu_0 < \kappa/\mathcal{C}(\kappa)$ , while equilibrium existence requires  $\mu_0 > \underline{\mu}$ . For  $\kappa < \kappa_{m,A}$ , these conditions are incompatible.<sup>21</sup> In Case B, the welfare effect is ambiguous and depends further on the equilibrium parameters.

**Lemma C.9** (Exhaustiveness). *Generically in  $(\mu_0, \delta)$ , the equilibria characterized in  $r_1^P - r_2^P$  and  $r_1^M - r_4^M$  exhaust the set of PBEs in which Buyer 1 acquires his private signal.*

*Proof.* The argument parallels the proof of Lemma C.5. In a pure-strategy PBE, the preceding characterization shows that Buyer 2 can follow only the two acquisition configurations underlying  $r_1^P$  and  $r_2^P$ . In each case, Buyer 2's continuation behavior uniquely determines the  $L$ -seller's continuation payoffs and hence her period-1 reservation price. Imposing the necessary condition  $p_1^* \leq v_L$  yields exactly the discount-factor restrictions defining  $r_1^P$  and  $r_2^P$ . Thus no additional pure-strategy PBE can arise.

For mixed-strategy PBEs, the preceding lemmas establish that the  $L$ -seller can mix only at  $p_1^* = v_L$  and that, generically, all sustainable randomization patterns are precisely those underlying  $r_1^M - r_4^M$ . The relevant indifference conditions and feasibility of the mixing probabilities yield exactly the parameter restrictions defining these four regions.

Therefore, apart from nongeneric boundary values, every PBE in which Buyer 1 acquires his private signal belongs to one of  $r_1^P - r_2^P$  or  $r_1^M - r_4^M$ .  $\square$

**Proof of Theorem 2** The proof follows from Lemma C.9 and preceding analyses of pure- and mixed-strategy equilibria. It only remains to define an appropriate  $\kappa^*$ . Let  $\kappa_m = \min\{\kappa_{m,A}, \kappa_{m,B}, \kappa_{m,C}, \kappa_{m,D}\}$  and  $\kappa_p = \min\{\kappa_{p,A}(\tilde{\delta}), \kappa_{p,B1}(\tilde{\delta}/\alpha), \kappa_{p,B2}(\tilde{\delta}/\alpha)\}$  as defined in the pure-strategy analysis. Since  $\kappa_{p,B2}(\tilde{\delta}/\alpha) = \kappa_{p,A}(\tilde{\delta})$ , equivalently,  $\kappa_p = \min\{\kappa_{p,A}(\tilde{\delta}), \kappa_{p,B1}(\tilde{\delta}/\alpha)\}$ .<sup>22</sup> Choose  $\kappa^{**} = \min\{\kappa_p, \kappa_m\}$ . For  $0 < \kappa < \kappa^{**}$ , the relevant information-cost constraints do not bind, meaning each equilibrium region is active wherever its respective requirement on  $\delta$  is satisfied. The welfare effects then follow as defined in the theorem.

<sup>21</sup>Here  $\mathcal{C}(\kappa) = \frac{(v_H - c_H)[(c_H - c_L)(3\alpha^2 - 3\alpha + 1) - \kappa(2\alpha - 1)] + (c_H - c_L)\kappa(1 - \alpha)}{\alpha(c_H - c_L) - \kappa} > 0$ . To see the incompatibility,  $\kappa/\mathcal{C}(\kappa) > \underline{\mu}^{p_1^*}$  is equivalent to  $P(\kappa) < 0$  for a quadratic  $P$ . If  $G(\kappa)$  is the quadratic whose smaller root is  $\kappa_{m,A}$ , then  $P(\kappa) = G(\kappa) + (c_H - c_L)\kappa^2 + \alpha(1 - \alpha)(v_H - c_H)(c_H - c_L)^2 > G(\kappa)$ . Since  $\kappa < \kappa_{m,A}$  implies  $G(\kappa) > 0$ , it follows that  $P(\kappa) > 0$ , and hence  $\kappa/\mathcal{C}(\kappa) < \underline{\mu}^{p_1^*} < \underline{\mu}$ .

<sup>22</sup>It is straightforward to verify that  $\kappa_{p,A}$ ,  $\kappa_{p,B1}$ , and  $\kappa_{p,B2}$  are decreasing in  $\delta$ .

### C.2.4 Proof of Lemma 4.3

Consider  $\mu_0 \in (\underline{\mu}^{p_1^*}, \underline{\mu})$ , with the bounds defined in (A.4). The equilibrium arguments are the same as in the common-acquisition region except for Buyer 1's acquisition and off-path constraints. I therefore record only the resulting modifications.

- A. (*Pure-strategy equilibrium*) Buyer 2 never acquires and offers  $c_L$  after trade and  $c_H$  after no trade. As in Case A of the pure-strategy proof, all of Buyer 2's equilibrium conditions are unchanged. Buyer 1's constraints become  $\underline{\mu}^{p_1^*} < \mu_0 < \underline{\mu}$  and  $\mu_0 < \frac{c_H - p_1^*}{v_H - p_1^*}$ . Hence, the profile is sustained for  $\underline{\mu}^{p_1^*} < \mu_0 < \underline{\mu}$  and  $\delta < \tilde{\delta}$ .
- B. (*Mixed-strategy equilibrium*) Buyer 2 mixes over acquisition after no trade and does not acquire after trade.

By Lemma C.7, the low-quality seller mixes at  $v_L$ . As in Case A of the mixed-strategy proof, Buyer 2 has two feasible on-path no-acquisition offer profiles.

- i.  $(p_2^n(t), p_2^n(nt)) = (c_L, c_H)$ .

The only changes are Buyer 1's constraints, which become  $\underline{\mu}^{p_1^*} < \mu_0 < \underline{\mu}$  and  $\mu_0 < \frac{c_H - p_1^*}{v_H - p_1^*}$ . Thus, this profile is sustained for  $\underline{\mu}^{p_1^*} < \mu_0 < \underline{\mu}$  and  $\tilde{\delta} < \delta < \tilde{\delta}/(1 - \alpha)$ .

- ii.  $(p_2^n(t), p_2^n(nt)) = (c_L, c_L)$ .

With  $p_1^n = c_H$ , Buyer 1's acquisition constraint is incompatible with Buyer 2's acquisition constraint after trade. Consider therefore the outcome-equivalent profile with  $p_1^n = v_L$ . Buyer 2's indifference after no trade yields  $\eta_{B2} = 1 - \frac{1-\alpha}{\alpha} \frac{\mu_0}{1-\mu_0} \frac{1-\underline{\mu}}{\underline{\mu}}$ , with  $0 < \eta_{B2} < 1$ . After trade, not acquiring and offering  $c_L$  requires  $\mu_2^t < \hat{\mu}$  and  $\alpha \mu_2^t (v_H - c_H) + (1 - \alpha)(1 - \mu_2^t)(v_L - c_H) + \alpha(1 - \mu_2^t)(v_L - c_L) - \kappa < (1 - \mu_2^t)(v_L - c_L)$ .

The low-quality seller's indifference condition,  $v_L - c_L = \delta \beta_{B2}(1 - \alpha)(c_H - c_L)$ , gives  $\beta_{B2} = \frac{v_L - c_L}{\delta(1 - \alpha)(c_H - c_L)}$ . Thus  $0 < \beta_{B2} < 1$  iff  $\delta > \tilde{\delta}/(1 - \alpha)$ . Together with Buyer 1's constraints, the profile is sustained iff  $\underline{\mu}^{p_1^*} < \mu_0 < \underline{\mu}$  and  $\delta > \tilde{\delta}/(1 - \alpha)$ .

**Welfare.** In this case, the welfare benchmark is  $ETS_{opaque}^n$  since the buyers do not acquire private signal in the opaque market. The expected social surplus corresponding to the equilibrium profiles above is given by (the subscripts denote the strategy profile under

consideration):

$$\begin{aligned}
ETS_A &= \mu_0 \left[ (\alpha + \delta(1 - \alpha))(v_H - c_H) - \kappa \right] + (1 - \mu_0) \left[ v_L - c_L - \kappa + \delta(v_L - c_L) \right] \\
ETS_{B1} &= \mu_0 \left[ (v_H - c_H) \left[ \alpha + \delta(1 - \alpha)(1 - \beta_{B1}(1 - \alpha)) \right] - \kappa [1 + \beta_{B1}\delta(1 - \alpha)] \right] \\
&\quad + (1 - \mu_0) \left[ (v_L - c_L) [1 + \delta - \alpha(1 - \eta_{B1})] - \kappa [1 + \alpha\delta\beta_{B1}(1 - \eta_{B1})] \right], \\
ETS_{B2} &= \mu_0 \left[ (v_H - c_H) \left( \alpha + \alpha(1 - \alpha)\delta\beta_{B2} \right) - \kappa \left( 1 + (1 - \alpha)\delta\beta_{B2} \right) \right] \\
&\quad + (1 - \mu_0) \left[ (v_L - c_L) \left( 1 + \delta - \alpha(1 - \eta_{B2}) \right) - \kappa \left( 1 + \alpha(1 - \eta_{B2})\delta\beta_{B2} \right) \right]
\end{aligned}$$

It can be verified algebraically that for  $\kappa < \min\{\alpha(v_H - c_H), (2\alpha - 1)(c_H - c_L)(v_H - c_H)/(v_H - v_L)\}$ ,  $ETS_A, ETS_{B1}, ETS_{B2} > ETS_{opaque}^n$ .

### C.2.5 Proof of Lemma 4.4

I construct an equilibrium supporting this outcome. Suppose Buyer 1 does not acquire information and offers  $c_H$ , which both seller types accept. Trade therefore occurs on path with probability one, so Buyer 2's posterior after trade is  $\mu_0$ . Since  $\mu_0 < \tilde{\mu}$ , Buyer 2 acquires information and offers  $c_H$  after a high signal and  $c_L$  after a low signal.

Following the off-path history of no trade, let  $\mu_2^{nt} = 1$ , so Buyer 2 does not acquire and offers  $c_H$ . The low-quality seller's indifference condition is  $p_1^* - c_L + \delta(1 - \alpha)(c_H - c_L) = \delta(c_H - c_L)$ , yielding  $p_1^* = c_L + \delta\alpha(c_H - c_L) > c_L$ . By Lemma 2, this implies  $\bar{\mu}^{p_1^*} < \tilde{\mu}$ , so Buyer 1 optimally does not acquire over the relevant interval.

The trade allocation is the same as under opacity, but transparency requires only Buyer 2, rather than both buyers, to acquire information. Hence information costs fall from  $2\kappa$  to  $\kappa$ , and transparency strictly raises welfare.  $\square$